The Diagnosis Agent Test Bed: Technical Companion

The method, the test bed, scoring and reproduction in full, for the archive of the diagnosis agent test bed

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1. Overview

This companion accompanies the paper “An Agent That Diagnoses the Optical Network: Measured Against Alarm-Based Troubleshooting” and the archive that holds its code and records. The paper reports what the agent does and what it achieves. The companion gives what a reader needs to check that account and reproduce it:

- the test bed, with the network drawn in full;
- every generation line and every winnowing tool;
- every setting of the ledger;
- the rules by which each case is scored and costed;
- every modelled quantity with its source;
- the steps that reproduce each result.

Alarm-based troubleshooting is the diagnostic approach that equipment vendors document and support. Alarms are correlated to a root alarm and its documented probable causes, and an engineer then follows the equipment’s troubleshooting procedures. The paper compares the agent with it at three points:

- **alarm correlation alone**;
- **manual troubleshooting** with the procedures;
- **manual troubleshooting from correlation**, which starts the same procedures from correlation’s root alarm.

1.1 What the archive holds

The archive holds:

- the world simulator, which plays the network and its faults;
- the knowledge files the agent reads;
- the agent itself: its tools, its planners and its ledger;
- the harness, which generates, scores and costs each case from its seed;
- the manifests of file hashes under which every arm ran;
- one record per case and arm.

An **arm** is one configuration of an approach run over a set of cases: the agent as reported, the agent with one tool switched off, manual troubleshooting, and so on. The code in the archive is the code frozen under the manifest C4H, and a check confirms that the files match it (Section 8).

1.2 The case sets

Every case is generated from its seed, so the same seed always gives the same network, fault, data and question. Four sets of seeds are used.

Table 1. The case sets

Set	Seeds	Use
Sealed cases	7000 to 7319	Drawn after the agent's code was frozen, examined by no one before, run once. Every headline result of the paper.
Diagnosis cases	2000 to 2319	Used in development; the analyses that need many arms or runs (the LLM deciding alone, telemetry capabilities, robustness, the degraded knowledge graph)
Completion test cases	3160 to 3319	Fresh cases for the test of documentation completion, never used before it
Generation cases	1000 to 1099	The study of the generation lines alone

Each set holds the eight kinds of anomaly in equal number:

- hard failures;
- soft degradations;
- transients;
- slow drifts, posed as early warning;
- faults of the monitoring itself;
- two coincident faults;
- faults after maintenance;
- cuts of a duct shared by several cables.

1.3 How the agent works, in brief

The agent has four parts: a knowledge graph, a set of tools, a planner and a ledger. A diagnosis runs in three steps.

1. Fixed queries build a **case frame**, the record of what is known before any cause is proposed (Section 3.1).

2. **Generation lines** then propose candidate causes, each with a prior probability (Section 3).

3. **Winnowing** gathers evidence until the ledger can commit or the budget is spent (Section 4).

When the evidence points to a cause not yet proposed, the agent returns to generation. The ledger is deterministic code that alone holds the probabilities and alone decides. A large language model (LLM) can take part as one generation line, as the planner of checks, as the reader of the operator's question and as the writer of the answer. In every role its output enters the agent as a proposal, a choice or a reading to be checked, never as a decision.

2. The test bed

A diagnosis agent cannot be judged on a real network's faults without the true cause of each, so it was measured in a simulated world where only the simulator knows the truth. This section describes that world: the network, the simulator and its physics, the gap between the world and the records, the

operating history and data, the knowledge graph and the documentation. Section 5 sets out how every case is scored.

2.1 The network

The reference network, named ref-national-24, is generated from a seeded design procedure. It has 24 sites with reconfigurable optical add-drop multiplexers (ROADMs) on a 2,000 by 1,000 km plane, with optical transport network (OTN) switching at 8 hub sites. The sites are joined by 41 links of 175 to 647 km. Each link carries two fibres, one per direction, over spans of 61 to 99 km: 336 fibre spans and 418 amplifiers in all, with 12 per cent of spans aerial. The band is the C-band, with 64 slots at 75 GHz.

As built, 225 coherent lightpaths carry 400G DP-16QAM (dual-polarisation 16-QAM, up to 800 km) or 200G DP-QPSK, with nominal margins of 2.0 to 9.6 dB. Over them run 400 client services (10GE, 100GE and 400GE), carried on paths of optical data units (ODU paths) of one to five hops through the hub switches. Of these, 116 are protected by subnetwork connection protection (SNCP) and four by OCh 1+1 protection, all planned from the plant records. At day 2190, when the cases are posed, 380 services are in service.

Ten shared-risk groups exist: eight ducts shared by links leaving a site in nearly the same direction and two long shared cables. Five of the ten are missing from the plant records, so protection planned from the records can be defeated by a cut of an unrecorded shared duct. Figure 1 draws the network.

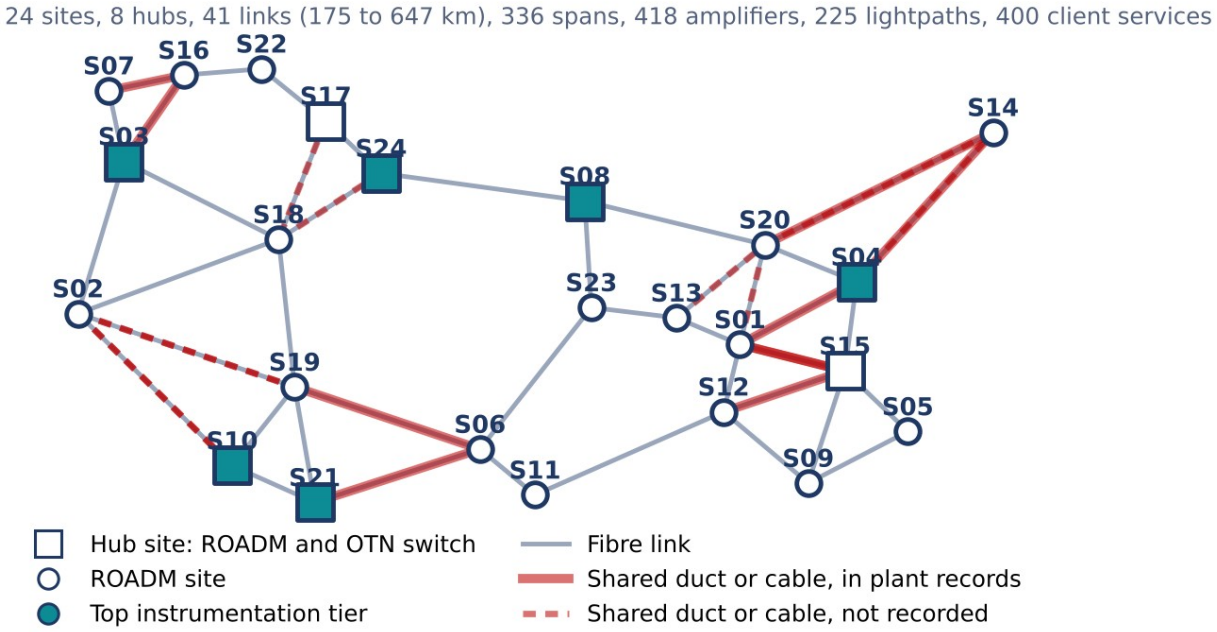


Figure 1. The reference network: 24 sites, 41 links, 336 spans, 418 amplifiers, 225 lightpaths and 400 client services as built. Route lengths are in Table 2. Hub sites carry an OTN switch; shared ducts and cables are drawn in red, dashed where the plant records omit them; filled sites have the top instrumentation tier.

Table 2. Route length of each fibre link

Link	km	Link	km	Link	km
S01 to S04	337	S04 to S20	252	S10 to S21	230
S01 to S12	176	S05 to S09	285	S11 to S12	524
S01 to S13	175	S05 to S15	222	S12 to S15	326

Link	km	Link	km	Link	km
S01 to S15	276	S06 to S11	182	S13 to S20	291
S01 to S20	261	S06 to S19	497	S13 to S23	218
S02 to S03	404	S06 to S21	442	S14 to S20	647
S02 to S10	550	S06 to S23	458	S16 to S22	198
S02 to S18	544	S07 to S16	195	S17 to S18	324
S02 to S19	582	S08 to S20	471	S17 to S22	210
S03 to S07	184	S08 to S23	271	S17 to S24	194
S03 to S16	267	S08 to S24	523	S18 to S19	377
S03 to S18	441	S09 to S12	281	S18 to S24	310
S04 to S14	515	S09 to S15	300	S19 to S21	298
S04 to S15	228	S10 to S19	255		

Equipment comes from two fictitious vendors, VA in the west and VB in the east. There are 26 equipment types, each with a **description package**: its documentation in the form the graph holds (Section 2.5).

2.2 The world simulator and its physics

The world simulator evaluates the network once per interval in which its state is constant. For every lightpath it computes amplified spontaneous emission and nonlinear interference, the power re-levelling of the wavelength-selective switch (WSS) at each ROADM, received power, optical signal-to-noise ratio (OSNR) and generalised SNR (GSNR), the bit error rate before forward error correction (pre-FEC BER) from the GSNR, chromatic dispersion, differential group delay (DGD) and the rate of change of the state of polarisation (SOP). The OTN layer evaluates each direction of each service over its ODU path, applies protection and raises the alarms an operator would see.

From these intervals the simulator writes alarms, protection-switch events, 15-minute performance counters and sampled instrument streams with realistic noise. A sub-second transient is therefore represented exactly even when instruments sample slowly.

Faults are drawn from a catalogue of 31 fault modes in six groups: hard failures, soft degradations, transients, incipient drifts, context (maintenance, provisioning changes and weather) and faults of the monitoring and records. Each mode has its physical effect in the simulator. The agent never sees the catalogue. It sees the vendor documentation, which describes most modes at the coarser grain of documentation and some not at all (Section 2.6).

2.3 The world, the records and the twin

The agent's twin (Section 4.4) is a predictive digital twin of optical transmission, built from the records alone: it can predict what the network would show in a state it is not in. It predicts optical powers, OSNR, bit error rate, DGD, polarisation, loss of signal, line errored seconds and the answers of measurement checks under any hypothesised fault. It does not predict equipment behaviour, such as the alarms raised or protection switching.

The gap. The world differs from the records by a designed **gap**. Each amplifier's true noise figure and gain set-point, each span's fibre attenuation, connector loss and polarisation-mode dispersion (PMD) and each

instrument's calibration offset are drawn once per network, while the records keep the datasheet and planned values. The gap is **calibrated**: its spread is centred on the datasheet values, as the residual error of a twin calibrated against the network's monitors would be. Its size is the nominal one, called scale 1: fibre attenuation about 0.20 dB/km (standard deviation 0.008), connector loss about 1.0 dB (standard deviation 0.25), noise figure about 5.5 dB (standard deviation 0.3), gain set-point error with a standard deviation of 0.3 dB, PMD log-normal and calibration offsets within ±0.5 dB.

The twin's error. From its inputs, the gap gives the twin an error of 0.16 dB rms (root mean square; measured over 450 receivers with no fault, mean -0.07 dB, standard deviation 0.14 dB, largest 0.97 dB) on predicted margin. The twin adds its own calculation error of 0.18 dB (one standard deviation), so that its error on predicted margin is about 0.24 dB in all. That matches what the authors' team finds in practice over very long amplifier chains: predicted margin within 1 dB in the worst case and almost always within 0.5 dB. From the gap alone its error on predicted OSNR is about 0.52 dB (one standard deviation). Measured the same way on the frozen code, its error from the gap is 0.60 dB rms on span loss (336 spans, mean -0.04 dB, largest 2.0 dB), 0.22 ps rms on a receiver's differential group delay (DGD), about a tenth of the 1.4 ps it predicts on average (mean 0.14 ps, standard deviation 0.17 ps), and 0.15 decades rms on the pre-FEC bit error rate (mean -0.06, standard deviation 0.14, largest 1.1 decades).

A twin with wrong physics. The paper's robustness test places the agent in a world that has physics the twin does not model: the world's nonlinear interference is 1.4 times the twin's, amplifier noise figures rise with age by up to 0.5 dB, the in-line amplifiers carry a gain tilt with a standard deviation of 0.5 dB and connector losses spread with a standard deviation of 0.5 dB, twice what the records state. Commissioning gains are set again to the new span losses, as turn-up would set them. The twin and the statement of its accuracy are unchanged. Measured on the frozen code, the mismatch puts the world's GSNR about 0.55 dB below the twin's prediction (standard deviation 0.15 dB): mostly a shift of the mean.

Records of poorer quality. The paper also tests the agent with its knowledge degraded while the world is untouched. Each degradation is one fixed set of errors, as an operator's records carry one set of errors:

- **inventory**: 5 per cent of equipment recorded with the wrong type of the same kind; 30 per cent of spans' cable and duct records, 10 per cent of protection records, 10 per cent of instrument records and 5 per cent of cross-connect records lost; the twin inherits these errors;
- **flat graph**: the layered dependency relations kept only as untyped, undirected connectivity, without order, ends or role; the twin keeps the true line design;
- **fault signatures**: for 30 per cent of pairs of equipment type and fault mode, no documented effects and no alarm naming the mode as a probable cause;
- **fault modes**: 20 per cent of pairs of equipment type and fault mode removed, never all of a type's modes.

2.4 The operating history and the data

The network comes with six simulated years of history: build-out, provisioning, the roll-out of instrumentation site by site and operations, with 7,442 incidents and 1,757 trouble tickets. Tickets record what operations would have recorded, not the truth. Of their explanations, 799 are accurate, 494 coarse, 148 wrong and 316 "no fault found". Records arrive with realistic lags, and four instrument deployments were never registered.

Instrumentation is deployed in tiers, site by site, and everything it reports is kept in the **data lake**, the store of every stream, alarm and counter.

- The baseline tier gives alarms, protection events and 15-minute counters.
- The power tier adds amplifier and receiver power every 10 seconds.
- The channel-monitor tier adds per-channel power and OSNR at each ROADM degree every minute.
- The coherent tier adds coherent receiver telemetry (pre-FEC bit error rate, an OSNR estimate, DGD and SOP rate every second) and the health streams (amplifier pump current and transmitter laser bias).
- The full tier adds triggered captures, context records and embedded optical time-domain reflectometry (OTDR).

All the diagnosis cases are posed at day 2190, when the network is mature and most sites are at the coherent or full tier.

Instrument noise. Every stream and every check answer carries the noise of its instrument, as Table 3 gives it. The agent never knows the draw; it allows for the instrument's stated accuracy (Table 8).

Table 3. Instrument noise (one standard deviation)

Source	Noise
Streams	Power 0.2 dB; OSNR 0.5 dB; DGD 0.3 ps; SOP rate 10 per cent plus a small positive offset (about 50 rad/s); pump current and laser bias 0.5 per cent of nominal; bit error rate as computed
Performance detail	Minimum received power 0.1 dB
Coherent diagnostics	OSNR 0.3 dB; bit error rate 0.1 decade; DGD 0.3 ps; SOP rate 10 per cent; margin 0.2 dB
Transmit power, power meter, span loss	0.1 dB
OTDR	Position 0.05 km; loss step 0.1 dB

Two instrumentation branches. The results are reported on two branches of instrumentation. **Status-quo instrumentation** is the deployment the network's history gives on day 2190: 18 of the 24 sites at the coherent tier and 6 at the full tier, with telemetry polled at fixed rates and 15-minute performance monitoring. The **future instrumentation** puts every site in service at the full tier, the top tier of the history, deployed and registered 30 days before the case. Over status-quo instrumentation it adds, at the 18 coherent sites, embedded OTDR, sub-second triggered capture at the receivers and telemetry of health and context. The further telemetry capabilities the paper measures, such as measurement liveness, are not part of the future instrumentation; each is added to the world on its own and measured there.

2.5 The knowledge graph

The graph is built by replaying the lifecycle events of the history up to the question's day. At day 2190 it holds 7,718 nodes and 21,880 edges. They describe:

- the equipment and its containment;
- the fibre plant, with its cables and ducts as recorded;
- the lightpaths, ODU paths, services and protection groups, with their signal-flow edges;
- 3,643 instrument nodes, with what each observes;

- the description packages: functional blocks, alarms and their probable causes, fault modes with their documented effects and natures (hard, soft, transient or incipient), monitoring points and the defect correlations of OTN functions defined in ITU-T Recommendation G.798;
- the troubleshooting procedures, where they are held in the graph rather than by the procedure tool;
- base rates of causes from closed tickets, weighted by how each was verified.

Completeness. Every relation carries a declaration of its completeness, which the tools respect. Equipment types, signal paths and protection are complete once recorded, so a missing fact may be read as false. Duct and cable membership and instrument registrations are incomplete, so a missing fact there is only unknown. A tool may reason from absence only over a complete relation.

2.6 Documentation and what it leaves out

The equipment is of two fictitious vendors. Their documentation (equipment types, fault modes, alarms and probable causes) and one set of alarm-clearing procedures serving both were written for the test bed, from ITU-T Recommendation G.798 and the form of typical vendor alarm and troubleshooting references, vendor-neutral and independently of the simulator. None of it is taken from any vendor's products or documentation, so the results say nothing about any real vendor's equipment or procedures. The description packages are written at the grain of documentation, and they differ between the two vendors in realistic ways. Vendor VB does not document the amplifier noise-figure rise, and only vendor VA documents transmitter laser-bias telemetry. Two transient modes of the world, the channel-loading transient and the protection-switch hit, are documented by neither vendor nor by the practice documents. They are there to test whether the agent can say that the cause lies outside what it knows. The twin's practice notes nevertheless give the twin a scenario for a channel-loading excursion, as a predictive twin can simulate one although no vendor documents it as a fault (Section 4.4).

3. The case frame and generation

Generation proposes the candidates that winnowing will judge, and no winnowing can recover a cause that is never proposed. This section describes generation in full. It begins with the case frame, on which every line depends, then defines what a candidate is and describes the two services that several lines share: grounding and forward propagation. It then takes each line in turn and sets out what it takes in, what it does, what it gives out and where it is weak. It ends with the language-model proposer, the pooling of the lines into one prior and the return passes that run generation again during winnowing. Section 4 describes winnowing.

3.1 The case frame

The frame is a record of what is known before anyone proposes a cause. It is built once per question by the **framer**, a deterministic program that runs fixed queries over the knowledge graph and the data lake. The frame names no cause. It records the following.

- **The question, its anchor and time.** The **anchor** is the service, element or customer the question names; the time is when the question was asked.
- **The onset and the time window.** The onset is estimated as the earlier of two times: the first alarm in the window and the first persistent departure of any stream in the anchor's scope. A departure must exceed eight robust scatters from the earlier median and persist for three samples (the robust scatter is defined below). The window opens 15 minutes before the question.

- **The scope.** It starts from the anchor's closure: for a service, its working and protection paths; for an element, the signal paths of the lightpaths through it and the services they carry. It is widened by the signal paths through every affected element, up to 800 elements.
- **The symptoms.** These are every alarm and protection-switch event in the window, every instrument stream in scope whose reading changed at onset and every new errored second anywhere in the network. Each symptom carries the identifier of the record behind it.
- **The services.** Each affected service's state is recorded as down, hit by a protection switch, errored, degraded or unaffected. For a service anchor, up to 30 unaffected neighbours are included too. What did not happen to a neighbour narrows the cause as much as what did.
- **The shared elements.** When several services are hurt, these are the elements on which all their working paths depend.
- **Observability.** The instruments in scope that reported and saw no change are evidence too. They are recorded stream by stream and, for channel monitors, channel by channel, so that a monitor examined on some channels counts as quiet only on those. The frame also records the streams that went silent and any readings from sources not on record.
- **Context.** Maintenance windows at sites in scope in the last seven days.

How a reading change is recognised. One generic detector tests every recorded stream in scope at the onset. Its unit of change is the **robust scatter**: the larger of 1.48 times the median absolute deviation and half the standard deviation, with a small floor, so that a few wild samples cannot inflate it. A stream is **lost** or **restored** when more than half its samples change state. It shows a **level** shift when its median after onset differs from the median before by more than four robust scatters and the shift is statistically significant ($|z| > 6$). It shows an **excursion** when its peak departs more than ten robust scatters from the earlier median and then recovers.

Large faults give large frames: a fibre cut on a busy line produces over a thousand symptoms. A drift has no onset in the window, so its frame shows no change at all. That is why the trend line exists (Section 3.11) and why two winnowing terms read the present level of slow-moving telemetry rather than its changes (Section 4.7).

Wider frames for return passes. The framer takes a widening factor. At factor 1 it builds the frame described above. A return to generation asks for a frame at factor 2, 4 or 8 (Section 3.15). Four quantities grow by the factor: the scope cap, the signal paths followed from each affected element, the lightpaths and services taken from the anchor and the streams examined to set the onset. In addition, the network elements adjacent in the graph to every affected element join the scope.

The LLM framer. As an alternative that the paper measures, the first frame can be built by a large language model (LLM) instead. The LLM has eleven query tools: list alarms, protection switches, new errored seconds, service information, element descriptions, signal paths, services through an element, common elements, reading changes, observability and maintenance. It has up to 24 turns, its lists are capped at 60 items and it is told not to propose causes. Its reading-change tool does not cover the per-channel monitors, which the deterministic framer reads slot by slot. Return passes widen the frame with the deterministic framer whichever framer built the first one.

3.2 Candidates and what every line returns

Hypotheses. A **hypothesis** is a documented fault mode at a location or a **compound** of several such causes occurring together. A location is one of four things: an element identifier; a ROADM channel,

written “oms|slot” (OMS is the optical multiplex section, the multiplexed signal between two ROADMs); a WSS input port, written “oms|port” (the port on which that OMS arrives at the ROADM where it ends); or a cross-connect, written “switch|ODU path”.

What every line returns. Every generation line takes the frame and the graph and returns the same thing. It gives a list of candidates, each with a **prior mass** (the share of the line’s probability it receives) and a rationale. It also gives a **residual mass**: the line’s probability that the cause is none of its candidates. A line that is sure of its candidates keeps a small residual, and a line that explains little of the frame keeps a large one. The same cause proposed by two lines has the same key and merges in the ledger.

The coded lines. Six general lines run without an LLM: the abduction walk, set cover, random-walk ranking, alarm correlation, analogy with past cases and the troubleshooting procedures. They name no fault mode in their code. Modes, effects, alarms and locations are all read from the graph, so these lines reason only from what the documentation says. Three narrow **specialised lines** cover kinds of anomaly for which the general lines have no starting point; unlike the general lines, they name in code the fault modes they look for. Two learned specialists and the LLM proposer complete the set. Table 4 lists every line, where it runs and whether the agent as reported in the paper uses it.

Table 4. The generation lines

Line	Kind	Runs in	Pooled weight	In the agent as reported
Abduction walk (3.5)	Coded, from documentation	Every pass	1; 0.25 in return passes	Yes
Set cover (3.6)	Coded, from documentation	Every pass	1; 0.25 in return passes	Yes
Random-walk ranking (3.7)	Coded, from documentation	First pass	1	Yes
Alarm correlation (3.8)	Coded, from documentation	First pass	1	Yes
Analogy with past cases (3.9)	Coded, from closed tickets	First pass	1	Yes
Troubleshooting procedures (3.10)	Coded, from the procedure library	First pass	1	Yes
Monitoring, shared-risk and trend lines (3.11)	Specialised, coded	First pass	0.25	Yes
Transient event recogniser (3.12)	Learned specialist	First pass	1	Yes
Ablative pairing and the two-leaders compound (3.15)	Coded, from the twin’s residual and the ledger	Return passes	0.25	Yes
Multiple-fault specialist (3.15)	Learned specialist	Return passes	0.25	No; measured separately
LLM proposer (3.13)	Language model, grounded	First pass	1	No; measured separately

How the lines are scored on their own. The main measure of a line on its own is **class top-1**: the share of cases in which the line's first candidate is the true cause or a cause the data cannot tell apart from it (a member of the truth's equivalence class, Section 5.2).

3.3 Grounding

Documentation and past tickets speak of locations in relative terms: "the span feeding the unit", "the far-end transmitter", "a switch on the ODU path". **Grounding** turns such a term into concrete elements of the network. It is a tool of its own, shared by the LLM proposer, analogy, alarm correlation and the procedures, so that no line reaches into another's internals.

It takes in a frame, a fault mode and a location. The location can be an element, a location term with an anchor (a symptom or a service), "shared" or a relation from a past case ("upstream of the amplifier that raised input loss").

It does three things.

1. It resolves the location against the graph and the case. It uses a vocabulary of 15 location terms, held in the graph in the way vendor documentation phrases its probable causes: this unit, this port, the upstream span, the upstream amplifier, the OMS input, the upstream ROADM channel, upstream on the signal's path, the far-end transmitter, the far-end receiver, the path to the far end, either port, a switch on the ODU path, and the working, protection and active ODU paths.

2. It expands coarse locations. A whole ROADM named for a per-channel or per-port fault becomes the case's channels at it and the WSS input ports on which they arrive. A section (OMS) becomes itself and the WSS input port on which it arrives. A whole OTN switch becomes the cross-connects of the case's hurt services.

3. It filters by documentation, at the documented grain. A location is kept only if its equipment documents the mode. A mode whose effect is per channel lies only at a ROADM channel. A mode whose effect falls on every channel arriving on one degree, such as a WSS port failure, lies only at a WSS input port. A per-ODU mode lies only at a cross-connect.

An option, the **documentation gap**, also keeps modes that another equipment type of the same class documents, at a reduced weight.

It gives out a list of locations with weights, 1 for documented ones.

Its limits. A correct proposal is lost when its location resolves to equipment that does not document the mode or when a reference is mis-resolved. Most often a proposal grounds to nothing because nothing at the named location documents the mode or because a site was named whose node equipment does not.

3.4 Forward propagation

Several lines and tools need to know what a fault would cause. The **propagator** answers that from the graph alone. Given a fault mode at an element, it reads:

- the mode's documented effects: what changes, at what scope, starting where and of what nature;
- each equipment type's signal behaviour: what passes from input to output, what a ROADM routes per channel and where a power change is absorbed within a margin;
- what each instrument sees and at what scope;
- each alarm's definition: when it is raised, how it is suppressed and what it is reported against;

- the defects, consequent actions and SNCP selection of the OTN functions defined in ITU-T Recommendation G.798.

It returns a **prediction**: for every observation the fault would produce (an alarm on an element, a change in a stream, a service down or hit), a strength. Hard faults give **required** effects, soft and incipient faults **expected** ones and transients **possible** ones. An uncertain transfer from one element to the next downgrades a strength to possible. A transient's prediction is flagged so that it cannot explain a lasting change. An effect documented at port scope, such as the signal loss of a failed WSS input port, starts at the ROADM on every channel arriving on that port and on no other.

Several causes can act together, as in a compound or the cut of a shared duct. The propagator then accumulates the optical and digital effects of every cause before it judges the services, so that protection is judged once. A working path hit by one cause and its protecting path hit by another therefore take the service down.

3.5 The abduction walk

The abduction walk proposes the causes that would best explain what was observed. It walks back from the symptoms to everything that could have produced them and then checks each candidate forward against everything that was examined.

It takes in the frame and the graph.

It does five things.

1. It turns the frame into **observations**: the alarms, reading changes and errored-second symptoms together with the services down or hit. It marks what was **examined**: every alarm, every service and every stream of an instrument that reported, whether changed or quiet.

2. It **walks backward** from each symptomatic element to everything upstream of it on the signal paths through it. For a service this is its whole closure; for a ROADM channel, the path up to that ROADM. Each upstream element gains support equal to the number of symptoms at the element the walk started from, so an element upstream of many symptoms gathers support from all of them. Every element with at least half the best support is kept, up to 400. ROADMs expand to the channels of the symptomatic lightpaths and switches to the cross-connects of hurt services.

3. For each element kept it takes every **documented fault mode** of its equipment, at the documented grain.

4. It **checks each candidate forward**. The propagator predicts what the fault would cause. The candidate is credited with each observation the prediction contains, except that a transient cannot explain a lasting change. It is debited 1 for each required effect that an examined instrument did not see and 0.3 for each such expected effect. Candidates that explain nothing are dropped.

5. It weighs each candidate by a **base-rate prior**: the documented failure rate on the fault-mode edge of the graph, scaled by how often the mode has been recorded on that equipment type in closed tickets, each ticket weighted by how its cause was verified.

It gives out the 25 best candidates. Each candidate's score, in log-odds, is one unit for each observation it explains, less its debits, plus half the logarithm of its base-rate prior. The scores are normalised into prior masses. The residual grows with the share of observations the best candidate leaves unexplained: 0.05 plus 0.4 times that share, at most 0.5.

Its limits. The forward check is what makes the walk abductive in the full sense: a candidate is credited only for what it would actually produce and pays for what it would produce that nobody saw. It is also where the cost lies, since every candidate is propagated through the graph; most of the graph queries are repeats, which caching removes. The walk ranks candidates by how much they explain, so it is weak where a fault leaves one subtle reading among louder candidates, as soft degradations often do. It is strong on transients, whose excursions and counters it can explain. It proposes only documented modes at the documented grain. A mode that no document attaches to an element cannot be proposed there by this line or by any other line built on the documentation.

3.6 Set cover

Set cover looks for the smallest set of causes that together explain the observations. Of the general lines, it is the only one that can propose coincident faults in the first pass.

It takes in the abduction walk's scored candidates.

It does a greedy cover over the 40 best. At each step it picks the candidate with the highest score, where the score is the number of observations it explains that are not yet explained, less its debits, plus a tenth of the logarithm of its base-rate prior. It stops when 95 per cent of observations are covered or three causes have been picked.

It gives out the first pick as a single cause, with mass 0.6 times the share of observations it explains plus 0.1. If more than one cause was needed, it also gives the compound of all picks, with mass 0.6 times the share left.

Its limits. It inherits the abduction walk's candidates and blind spots.

3.7 Random-walk ranking

Random-walk ranking ranks elements by how often a random walk upstream from the symptoms visits them. It has no forward check, by design: it is there to show what the forward check is worth.

It takes in the frame's symptomatic elements with their symptom counts.

It does a personalised PageRank on a graph of the signal paths through the symptomatic elements. The edges are reversed so that the walk runs upstream. At each step the walk continues with probability 0.85 (the damping) and otherwise restarts at a symptom, chosen in proportion to the symptom counts. The line then splits the visit probability of each of the 60 most visited elements over the element's documented modes, in proportion to their base-rate priors.

It gives out the 25 best pairs of mode and location with 0.9 of the mass. Its residual is fixed at 0.1.

Its limits. It drifts towards elements that many paths share, such as transmitters and ROADMs, so its first candidate is right far less often than that of the lines with a forward check. It does well only where the other lines fail on a single changed reading.

3.8 Alarm correlation

Alarm correlation reproduces what a network management system's correlation engine does, the first step of alarm-based troubleshooting. It is the step the agent's generation would replace, and it is kept as a line so that its proposals are weighed like any other.

It takes in the frame's alarms only. It assumes, as real equipment does, that each element has already applied the G.798 defect correlations among its own defects.

It does four things.

1. **Topological correlation.** It walks the graph over the alarms. One alarm explains another in three situations: when its element lies upstream of the other's on a signal path they share; when it is an equipment alarm of a unit and the other is raised on the same unit; or when the other is a client alarm of a service whose path passes through its element.

2. **Root alarms.** An alarm that no other alarm explains is a **root**. If every alarm explains another, as both directions of a cut do, all are roots.

3. **The root chosen.** It takes the root that explains the most other alarms, breaking ties by severity.

4. **The lookup.** It reads that alarm's documented probable causes and grounds each one onto the network.

It gives out the root alarm's grounded causes with equal mass (0.9 in all) and a residual of 0.1, as an engineer would receive them from the management system: a list, unranked. With no alarm it has nothing to say and abstains.

Its limits. It sees nothing without an alarm, so it cannot propose a cause for a fault that raises none. It also fails where the root alarm's documented causes point elsewhere. On a fault at an OMS input it proposes nothing useful, because the root alarm's documentation points to the span or the upstream amplifier, not to the patch. In effect, alarm correlation is the abduction walk restricted to alarms, to one root and to the vendor's list, with no forward check.

3.9 Analogy with past cases

Analogy proposes the causes recorded for similar past incidents. It is the one line whose performance grows with the network's history.

The experience store. Each closed ticket becomes a **past case** when the ticket closes. The past case is written with the graph as it then stood and revised only by dated follow-up findings. It holds two things. Its **role signature** describes what was seen: the alarm types with the class and role of the element that raised each; the numbers of services down, switched and degraded; and whether any alarm was raised. Its recorded cause is held as a **relation** to what was seen ("the span upstream of the amplifier that raised input loss"). The store holds 306 past cases at day 730 and 1,753 at day 2190. Each is weighted by how its explanation was verified: 1.0 when a repair confirmed it, 0.7 when an engineer concluded it, 0.3 when the agent concluded it without confirmation and 0 when it was refuted.

It takes in the frame and the store as of the question's day.

It does a retrieval of the 20 past cases most similar to the frame, by weighted overlap of signatures. Each votes for its cause with a weight equal to its similarity times its verification weight. The cause's relation is grounded onto the current network.

It gives out the 25 best grounded causes. The residual is the share of votes that could not be used (no fault found, no usable cause or a cause that could not be grounded), plus 0.05.

Its limits. It cannot propose a fault that has never been ticketed, so it says nothing useful about OTN switch faults or transients. It is weak where tickets were coarse. It improves as the store grows.

3.10 Troubleshooting procedures

The procedures line does from the office what an engineer following the vendors' alarm-clearing procedures would do.

The library. It holds fifteen vendor-neutral alarm-clearing procedures, written in the form of vendor procedures, with an order for choosing among them. The same content exists in two forms: held by the tool as its own documents or published into the graph by a lifecycle event and read back through general queries.

It takes in the frame and the data held.

It does the following. It starts from the first applicable entry of the selection order: an alarm present; failing that, a degraded receiver, an errored client port or the receivers on the path of the question's service. It then walks the procedure's decision tree. A step that asks about an alarm is answered from the alarm records, a step about a reading by the change detector at the onset and a step about a unit's role from the graph. A field test it cannot perform is recorded as advice, and both of its outcomes are followed.

It gives out an equal share of 0.9 to each leaf of the procedure that is reached, split over the leaf's causes and their grounded locations.

Its limits. It is weak on spans: a fibre cut is held open against added loss until an OTDR trace could decide. It is also weak on transients and on cuts of both fibres of a cable, and it has no starting point for faults without an alarm or a degraded receiver. The procedures were tuned on development cases, so their scores on those cases are not a fair measure, and only scores on other cases are reported.

3.11 The specialised lines

The diagnosis cases include kinds of anomaly for which the general lines have no suitable starting point: drifts, which show no change at onset; faults of the monitoring, where the evidence itself is wrong; and duct cuts, which take down several links at once. Three narrow lines, designed on development cases, cover them. They run in the first pass of generation only.

The monitoring line. It takes the elements with a changed reading but no alarm of their own, together with any amplifier or line-interface anchor. For each such element it takes its power instrument, at most six instruments in all. For each instrument it proposes two hypotheses: that the instrument misreports by a fixed bias and that its telemetry has frozen. Without it, no line would ever suspect the evidence itself.

The shared-risk line. It takes the lines that lost their input at once (amplifier input and OMS loss-of-signal alarms) and groups them by link. For each site at an end of two or more lost links, it proposes a compound: the cut of the span of each link nearest that site, as a dig-up in a shared duct or trench would cause. The compound gets mass 0.4 when the plant records place those spans in a common duct and 0.2 when they do not, because shared-risk records are known to be incomplete. The lower mass for spans not on record as sharing a duct is what lets the agent find a shared risk that the operator's records do not know.

The trend line. It takes the anchor elements that show no change at onset, plus up to five elements whose pump current or laser bias reads above 103 per cent of nominal. For each it proposes the documented ageing modes of its equipment, at double weight where its health level is high. Where an equipment type documents no ageing mode, it borrows those documented for another type of the same class, at half weight. An ageing mode whose documented effects only a health stream (pump current or laser bias) can see is proposed only at an element that reports that stream, since nothing else could judge it. Drifts have no onset, so without this line no candidate would be proposed for them at all.

The fibre-cut name. A cut of a cable takes down both fibres of a span, one in each direction. Before candidates enter the ledger, a cut at a span is given one canonical name for the cable section. The two fibres' candidates therefore merge instead of splitting the probability.

3.12 The transient event recogniser

Some events are brief and leave no lasting trace: a protection-switch hit, a channel-loading transient, a mechanical disturbance of a span and a polarisation (SOP) transient are over before the question is asked. Two of them, the channel-loading transient and the protection-switch hit, are documented nowhere (Section 2.6), so no line built on the documentation can propose them. The **transient event recogniser** is a learned specialist for these four kinds of event. It is a graph neural network (GNN): a model that reads the readings of the network's elements together with the way those elements are connected.

It takes in a graph of the network built from the inventory. Its nodes fall into four groups (OMSs; spans and amplifiers; receivers; lightpaths), joined by eight types of edge along the signal path. Each node carries features computed from its telemetry around the estimated onset, in a window that opens 150 seconds before it.

It does message passing between the node groups and reads the result with three heads: the kind of event (or none), its location and the logarithm of its duration. The location is restricted to elements that can host that kind of event. Its probabilities are calibrated by a temperature fitted on held-out cases of its own world. It **abstains** when the probability of "no transient" exceeds a threshold, set so that it abstains on at least 98 per cent of held-out cases with no transient.

It is trained on a different world. The training set is 8,000 cases on 234 other reference networks of 8 to 16 sites. It never includes the test network or any test case. The world is varied on purpose: fault parameters are scaled by up to $\exp(\pm 0.4)$, instrument noise by 0.5 to 2 times, monitoring tiers are drawn at random per site and the onset estimate is shifted by −120 to +60 seconds, as the framer's estimates are.

It gives out the event it recognises as a candidate at the location it places. A polarisation transient is proposed at every span of the OMS it crossed, which is the resolution the data allow. The recogniser runs in the first pass of generation and is pooled at full weight. When it abstains it proposes nothing and so is not pooled (Section 3.14).

Its limits. It proposes; it does not weigh evidence, and its candidates are judged in winnowing like any other. The twin's practice notes give a scenario for a channel-loading excursion on an OMS, so the twin can confirm or refute a candidate of that kind. A protection-switch hit is equipment behaviour, which the transmission twin cannot judge and the documentation does not describe (Section 4.13).

3.13 The LLM proposer

The LLM proposer asks an external language model to propose causes from a description of the case. It is an optional line. The agent as reported in the paper does not use it; the paper measures what it adds, both as a line on its own and pooled with the coded lines.

The call. The proposer makes one structured call per case to the external model, with no tools, through the provider's structured-output interface. The LLM is gpt-5.6, which the provider resolves to its strongest tier, Sol. No temperature or seed is set. The prompt is fixed and recorded with its version in the code. Four of its design points matter for grounding: it states which element "shared" names; it marks the question's service; it lets a site stand for its ROADM or switch; and it labels symptoms O1, O2 and so on, so that they cannot be confused with the site names S01 to S24.

What it is shown. The system prompt gives the LLM five instructions: propose faults that could explain the case; reason from what was not seen, since a quiet instrument that should have seen an effect is evidence against a cause; prefer the location that explains most; include less likely alternatives; and do not invent elements, modes or terms. The message has four parts.

1. **The case in role form.** This gives the question with its onset and time. It lists each service (V1, V2 and so on) with its state, protection and description. It lists each symptom (O1, O2 and so on) with its time, kind, observable, detail and element, together with the element's role in words ("inline amplifier 3 of 5 on the line S17 to S18"). At most 150 symptoms are listed, with a count of the rest. It also gives the shared elements with their roles, the quiet instruments grouped by element (at most 150 elements), up to 20 silent streams and recent maintenance.

2. **Documentation.** Every equipment type present in the case, each with its documented fault modes, their nature, description and effects, followed by the meaning of each effect.

3. **The location vocabulary.** The 15 terms of Section 3.3, with their meanings.

4. **The procedures**, as an option. These are the procedures the graph links to the case's alarms, introduced as a guide and not as a verdict.

What it returns. The answer follows a strict schema. It holds an assessment in words, a list of proposals and the probability that the cause is none of these. Each proposal gives a fault mode (restricted to the modes the case's equipment documents), a location (a term with an anchor symptom or service, "shared" with a reference or an element), a probability and a rationale that includes what argues against it. In the generation study the LLM gave a mean of 3.6 proposals a case, left a median of 0.03 of its probability unassigned to them (mean 0.07), and its probabilities with the unassigned mass summed to about 1.0.

How the answer becomes candidates. Probabilities are clipped to between 0 and 1 and scaled down if they sum to more than one; a shortfall goes to the residual. Each proposal is grounded (Section 3.3), and its probability is split equally over its locations. A proposal that grounds nowhere sends its probability to the residual, with the reason recorded. Candidates are not capped in number: one proposal of a broad term can ground to over 300 locations. When the proposer is used in the agent, it runs in the first pass and joins the six general lines at the same weight.

Where it could add. The proposer reads the same documentation as the coded lines, so where that documentation is complete it has no knowledge they lack. It can add where the documentation names a fault mode but gives none of its effects. The coded lines then cannot connect the symptoms to the mode, while the LLM can propose it from what such a fault does, and grounding accepts it because the mode is documented. Where a mode is missing altogether, grounding rejects the proposal too.

Its drawbacks.

- **Time and tokens.** In the generation study a proposer call took a mean of 15 seconds, with about 11,400 input and 1,230 output tokens a case, about 800 of them reasoning. The LLM framer, where it is used, takes a mean of 6.5 calls, 66 seconds and about 49,500 input and 6,800 output tokens a case (80 sealed cases). The coded lines take a few milliseconds a case at the median and the abduction walk about two seconds, up to 24 on the largest frames; they cost nothing beyond computing time. The proposer's cost is paid on every case, including the many that the data held settle without it.
- **Not deterministic.** No temperature or seed is set, so repeated runs can give different proposals. Stability across repeats is not reported.
- **What it is shown is the LLM's whole world.** What the frame omits or the caps cut, the LLM cannot see.
- **The hand-off through grounding.** A correct answer can be lost through an ambiguous reference or a location whose equipment does not document the mode. A broad term spreads probability thinly.

- **Documented modes only.** The schema restricts it to documented modes. An undocumented mechanism can reach the ledger only through the residual.
- **Calibration.** Probabilities are **calibrated** when they match how often the candidates are right. The proposer's probabilities are well calibrated when high: candidates it put between 0.80 and 0.95 were the truth in 97 per cent of cases, and those above 0.95 in all. In the middle they are too low, 0.31 stated against 0.50 observed. Its stated probability that the cause is none of its proposals averaged 0.06, against 0.05 of cases in which it did not propose the truth.
- **Weak on transients and spans,** where the walks are stronger.

The proposal as a rule. A proposal is in effect a rule in a fixed language: a documented fault mode, a location term and an anchor, which grounding evaluates against the graph. "Rule Based Reasoning for Agentic Fault Diagnosis" proposes an alternative, in which the LLM writes rules from retrieved documentation and the rules are validated before an engine evaluates them. That is more general, since it can express a mechanism that no documentation lists. Validation, however, shows only that a rule can be evaluated, not that it describes the equipment. Here the documentation is formalised once and the LLM chooses within it. That excludes errors of rule construction, at the price of reach.

3.14 Pooling

The ledger pools the lines into one prior as a weighted mixture. A hypothesis's mixture is the weighted sum of the masses the lines give it, a line that did not propose it counting zero. One per cent of the total is then spread evenly over every named hypothesis, so that a candidate only one line proposed is not driven to zero. **"None of these"**, the hypothesis that the true cause is something no line proposed, gets the weighted sum of the lines' residuals. Compounds carry the mass of the line that proposed them.

The six general lines and the transient event recogniser have weight 1, and the specialised lines 0.25. Two further rules apply. A line that proposes nothing **abstains**: it is not pooled at all. A **narrow** line (a specialised line or any line in a return pass) has its residual set to zero and its candidates renormalised before pooling, because it speaks only of its own kind of cause and must not count as a vote for "none of these". Without both rules, empty lines and return passes would inflate "none of these", and every return would raise the very mass that triggered it.

The mixture is a rule for pooling opinions, not a joint probability model. All the lines read the same frame, so their agreement is not independent evidence, and the prior is not trusted to be calibrated. That is why winnowing, not the prior, decides (Section 4).

3.15 Return passes

A **return to generation** runs a further pass of generation during winnowing, at most three in a case. It is triggered in two ways in the agent as reported: when the ledger signals that the candidates are failing (Section 4.2) and when the leader would commit but the residual test finds that it leaves too much unexplained (Section 4.5). With the LLM orchestrator as planner, the planner may also ask for one (Section 4.11).

A return pass works from a wider frame, built for the question at a widening factor of 2, 4 and 8 for the second, third and fourth passes (Section 3.1), so that symptoms outside the first frame can be explained. On that frame it does four things.

1. It runs the abduction walk again, with the cap on candidate elements doubled for each pass (800, 1,600 and 3,200) and the 40 best kept instead of 25. It skips every hypothesis that a check or a consistency rule has already contradicted.

2. It runs set cover again on the new candidates.

3. It runs the **ablative pairing**: the twin's fitted prediction of the leader is subtracted from what was observed, and the leader is paired with each of up to six causes proposed where the remainder points (Section 4.5).

4. It forms a compound of the two leading single causes when each explains observations the other does not, provided the first holds at least 0.2 of the probability.

The condition on the last step guards against a design pitfall found and fixed. Pairing a weak leader with another stacks one guess on another, and a compound so built could reach the commitment threshold while "none of these" still held most of the probability. The ablative pairing has no such condition, because restricting it to strong leaders lost true pairs of coincident faults.

All four join the mixture at weight 0.25, with no residual. Graph consistency and the documented equipment term are rebuilt on the wider frame and judge every hypothesis again, old and new. Every other term already in the ledger (the twin, the health and margin levels, every earlier check) is applied to the new candidates before the ledger may commit, so a late arrival is judged on the same evidence as the incumbents.

The backward walk's threshold of half the best support is the same in every pass. A wider frame therefore adds candidates only where its new symptoms give new elements enough support. The LLM proposer is not used in return passes.

The multiple-fault specialist. As an option that the paper measures, a return pass can also run the **multiple-fault specialist**, which proposes a second cause directly rather than where the residual points. It is a graph neural network that reads the twin's residual under the fitted leader, stream by stream in units of the stated combined accuracy of instrument and twin, together with the change features of each element. It names the element and fault class most likely to account for what the leader leaves unexplained or "no second fault". The agent pairs the leader with each of its top five proposals as compound candidates, with priors from its calibrated probabilities, and the twin then fits each pair jointly. It does not commit and adds no evidence term. It abstains when no compared reading is left unexplained beyond the combined accuracy and when its probability of "no second fault" exceeds a threshold set so that it stays silent on at least 98 per cent of single-fault cases of its own world. Like the transient event recogniser, it was trained only on generated networks of a deliberately different world, never on the reference network: 7,732 training cases, 5,429 of them with two faults. Its weakest classes are DGD rises and WSS filter shifts, whose residual is shared with their neighbours along a path.

4. The ledger and winnowing

Winnowing narrows the candidates of generation by weighing evidence for and against them. Generation proposes; winnowing decides which candidate is right, and its strength is what keeps the agent's verdicts safe from a poor prior. This section describes winnowing in full. It begins with the ledger, which holds every probability and alone decides. It then sets out the ledger's rules for committing, for saying "I don't know" and for sending the agent back to generation. It then describes each tool that scores the evidence already held (the twin, graph consistency, the documented equipment term and the health and margin levels) and the twin's ablative use. It ends with the checks, the interpreters that score their answers, the

planners that choose them and the loop that ties everything together. A last section sets out what the architecture asks of the twin.

4.1 The ledger

The ledger is where every number attached to a hypothesis is held, and it is the only part of the agent that decides. It is deterministic code; no language model writes to it. Generation lines, tools and planners propose candidates and supply evidence. The ledger alone turns them into a **posterior**, the probability of each hypothesis given all the evidence so far, and the ledger alone decides when a candidate is probable enough to act on. Its arithmetic is fixed and inspectable, and every term it holds can be traced to the tool and the evidence behind it.

Hypotheses. A hypothesis takes one of three forms: a single cause; a compound of several causes occurring together; or "none of these", which holds the probability that the true cause is something no line has proposed. Each hypothesis has a key built from its causes, so the same cause proposed by several lines is one entry. Instrument faults are single causes like any other: their location is an instrument and their mode is a misreport or frozen telemetry.

The prior. The prior is the pooled mixture of the generation lines (Section 3.14). The mixture is a rule for pooling opinions, not a joint probability model. All the lines read the same frame, so their agreement is not independent evidence, and the prior is not assumed to be calibrated.

Evidence terms. A tool supplies evidence to the ledger as a **likelihood term**. For each hypothesis it can judge, the term gives the logarithm of how probable the evidence would be if that hypothesis were true. For hypotheses it cannot judge, it declines. It may also give a **background** value, its judgement of "none of these". Log-likelihoods are measured in **nats**, the natural unit: a difference of 1 nat between two hypotheses is a likelihood ratio of about 2.7 to 1, and 4.6 nats is 100 to 1. The posterior is the prior times the product of the terms' likelihoods, normalised to sum to one. Four safeguards apply.

- **Clipping.** A term's values are clipped to at most 8 nats (a likelihood ratio of about 3,000 to 1) below the best value it gives, the background included. No single term can therefore eliminate a hypothesis beyond recovery. One consequence should be stated plainly: since one term can separate two hypotheses by at most 8 nats before tempering, a twin misfit of hundreds of nats counts the same as one of 8.
- **Tempering.** Each term is multiplied by a weight, its **tau**, which expresses how far its tool is trusted.
- **Sharing of evidence.** Terms that read the same evidence must not count it twice. Each term names the evidence items it read, and for every item the taus of the terms that read it must sum to at most one. Where they do not, each such term's tau is reduced to one over their number.
- **Neutrality.** A hypothesis that a term declined, did not cover or did not yet exist for receives the term's average value, so that it is neither favoured nor penalised. "None of these" receives the background if the tool gives one and the average otherwise.

How the evidence is shared. The data already held (every stream, alarm and counter in the data lake up to the question) form two evidence items. The **transmission evidence** is the readings of optical power, OSNR, bit error rate, DGD and polarisation, together with the errored-second counts of line ports. The **equipment evidence** is the alarms raised and the services' states (down or switched to protection). Three terms read these items, as Table 5 shows. The health levels, the margin levels and each check's answer are evidence items of their own.

Table 5. How the terms share the evidence

Term	Evidence read	Tau
Twin (C.4)	Transmission evidence	0.7
Documented equipment term (C.6)	Equipment evidence	0.7
Graph consistency (C.6)	Both, since its rules test changes in readings as well as alarms and services	0.3 on each
Health levels (C.7)	Pump-current and laser-bias levels	1.0
Margin levels (C.7)	Pre-onset bit error rate of each receiver	1.0
Each check (C.9)	That check's answer	1.0

The taus on each item of the data held therefore sum to one: twin and graph consistency on the transmission evidence, graph consistency and the documented equipment term on the equipment evidence.

Coverage. When a term is added, its targets are every hypothesis then active. A hypothesis that a term has neither scored nor declined is a **coverage gap** for that term. Gaps matter for the twin, because it is fitted only to the leading candidates, as they come to lead (Section 4.4). The ledger may therefore not commit while the leader or the runner-up still has a gap in any term.

4.2 Commitment, "I don't know" and the signals for going back

Commitment. The ledger commits to the leading hypothesis when four conditions hold together:

- it is a named hypothesis, not "none of these";
- its probability is at least 0.80;
- "none of these" is below 0.20;
- neither it nor the runner-up has a coverage gap.

Before the commitment stands, the loop runs the residual test on the leader (Section 4.5). If the leader leaves too much unexplained, the commitment is deferred and the agent returns to generation.

"I don't know". When the ledger does not commit, its decision is a **set**: the hypotheses, in decreasing probability, that together hold 90 per cent of the probability. When "none of these" is at 0.20 or more, the set is flagged as **cause probably not considered**. This set is the agent's "I don't know", and it states what is known: the causes still in contention with their probabilities, whether the true cause was probably never proposed and, from the planner's record, why no further check was made. An agent that is neither built nor scored to force a single cause can say this honestly. Whether it was right to say it is judged by the harness (Section 5.4).

Commitment at a coarser grain. The candidates in a set may still agree on where the trouble is or on what kind of trouble it is. Three contenders at one receiver (a slow loss of margin, a misreporting power reading and a frozen one) say "the trouble is at this receiver" with near certainty, though not which of the

three it is. An operator can act on that answer. At the end of the case, when the decision is a set and "none of these" is below 0.20, the posterior is therefore summed over successively coarser grains:

1. a single cause shared by the candidates, as when the contenders are one cause alone and that cause with various possible companions;

2. the element each candidate is about (an instrument counts as the element it measures, and a WSS channel or port as its ROADM);

3. the OMS, for an element on one;

4. the site;

5. the fault mode, wherever placed.

A candidate counts towards a place only if all its causes are at that place, so a compound spread over two places supports neither alone. The finest grain whose mass reaches 0.80 is reported as a **commitment at a coarser grain**. A shared cause is committed to in this way only when its single-cause candidate alone holds at least 0.2. That condition guards against a design pitfall found and fixed: without it, a cause shared by every member of a set could be committed to although the cause on its own was negligible. The ledger's own decision is unchanged, and the coarser commitment is reported beside it. It has no effect on the checks chosen or the effort spent, and it is scored as its own outcome (Section 5.3).

Signals for going back. The ledger also reports when the hypotheses in hand are failing, so that the agent can look for new ones. It signals **regenerate** in two cases: when "none of these" exceeds 0.25, or when the last three terms together reduced the posterior's entropy by less than 0.1 bit (a **stall**). It signals **misfit** when, in some term with a background, every named hypothesis scores at least 8 nats below the background. The loop answers either signal with a return to generation (Section 4.12).

Settings. The prior floor (0.01), the cap (8 nats), the commitment threshold (0.80), the "none of these" thresholds (0.20 to commit, 0.25 to regenerate), the set mass (0.90), the stall test (0.1 bit over three terms) and the condition on a shared cause (0.2) are fixed before the cases on which they are measured are drawn, and they are the same in every configuration.

4.3 Principles common to every winnowing tool

A winnowing tool joins the set only when four things are fixed: how it feeds the ledger, what data it reads, what it outputs and how what it adds is measured. Every check result is one of three verdicts for each hypothesis: supported, contradicted or unresolved. A check that fails or is unavailable leaves every hypothesis unresolved, because a missing measurement is not evidence that the measured condition is false. These rules follow "Rule Based Reasoning for Agentic Fault Diagnosis".

4.4 The twin

The twin is the agent's main source of evidence from the data held and its means of predicting what a check would show.

What it is. The twin is a predictive transmission model of the network. The agent uses it counterfactually: for each candidate cause it asks "if it were this cause, what would the instruments have shown?" and compares the answer with what they did show. Its physics engine is the world simulator's own transmission model, run on a network rebuilt from the records alone. The twin therefore has the world's physics, and its error comes only from what the records do not tell it and from its own calculation error. The paper tests a world with physics that the twin does not model.

What it predicts. The twin predicts transmission. That covers optical powers at amplifiers, receivers and channel monitors; OSNR; the pre-FEC bit error rate (the rate before forward error correction); DGD; the rate of change of the state of polarisation (the SOP rate); loss of signal; and the errored seconds of line ports, which follow the bit error rate. It also predicts the answers of the checks that measure these quantities: coherent diagnostics, performance detail, transmit power, the power meter, span loss and both OTDR traces, embedded and portable (Section 4.8).

What it does not predict. The twin does not predict equipment behaviour: the alarms raised, protection switching, client-side counts or the answers of equipment checks (connector inspection and cleaning, the cross-connect audit, tandem connection monitoring, loopback, spare-receiver retune and unit swap). The health streams (pump current and laser bias) lie outside it as well; the health term reads them (Section 4.7).

This is the scope of an operator's transmission twin. It predicts optical transmission performance under arbitrary scenarios, so it can predict any measurement of transmission. Which equipment alarm trips, its soak time and suppression and the response of a given release of the protection logic are behaviours of the vendor's implementation, which such a twin does not hold. The agent judges equipment behaviour from the vendors' documentation instead, in three places: graph consistency and the documented equipment term (Section 4.6), the documented meaning of each check (Section 4.9) and the documented propagation of the leader's causes in the residual test (Section 4.5).

What it knows and does not know. The twin is built from the knowledge graph, so it knows what the records hold:

- sites, links and spans, with their recorded lengths and nominal losses;
- amplifiers, with the datasheet's typical noise figure and the gain set at commissioning;
- line interfaces, with their modulation and the design rules' transmit power and transceiver SNR;
- lightpaths, ODU paths, services, protection and the instrumentation tier of each site, as recorded;
- launch power, drop target and channel loading, from the design rules.

It does not know the true fibre attenuation, connector and splice losses, the true noise figures and gain errors, the instruments' calibration offsets, the PMD of each span, changes not yet recorded, unregistered instruments or unrecorded shared ducts. With complete records and no gap between world and records, its inputs would be exact. In the test bed the world differs from the records by the calibrated gap of Section 2.3, which gives an error of 0.16 dB rms on the predicted margin of a lightpath and at most 1.0 dB (0.97 dB, the largest over 450 receivers) over every receiver of the network.

Its own calculation error. A real twin also errs in its calculations. The test bed's twin computes the world's own physics, so it is given a calculation error of its own, drawn from its own random stream. The error has three parts:

- an offset on every predicted power, OSNR, margin and span loss, with a standard deviation of 0.18 dB;
- a shift of the predicted bit error rate by half a decade per dB of that offset;
- an error of 5 per cent (one standard deviation) on every predicted change, DGD included.

Each part is fixed per quantity and element for the case, because the twin's error repeats each time the twin is asked: it does not average out over repeated predictions. Input and calculation error together give about 0.24 dB (one standard deviation) on predicted margin. That matches practice: over very long

amplifier chains, the author's team finds predicted margin within 1 dB in the worst case and almost always within 0.5 dB, both sources of error included.

Noise-free predictions. The twin predicts what an instrument would read without noise: the instrument's true reading in the twin's network, plus the twin's calculation error. The world's instrument noise is never the twin's to know. A twin that predicted with the world's own draw of noise would share the errors of the reading it is compared with, and it would appear more accurate than any real twin can be. The instrument's noise is instead allowed for in every comparison (Section 4.9).

Its stated accuracy. The twin's accuracy is stated in its practice notes, and the agent allows for exactly what is stated. On predicted margin the statement is 0.25 dB (one standard deviation). On a predicted change of a stream it is a floor plus a fraction of the change: 0.25 dB plus 15 per cent for power; 0.4 dB plus 20 per cent for OSNR; 0.3 decade plus 20 per cent for the log bit error rate; 1 ps plus 20 per cent for DGD. The same document states, field by field, the twin's absolute accuracy on the answers of checks, which enters the accuracy standard (Table 7). The statement is the same in every configuration. The paper's tests of a twin with wrong physics therefore measure what happens when the twin is worse than the operator believes.

Scenarios from documentation. To simulate a cause, the twin needs the physical change the cause makes. It looks up the scenario that the vendor documentation or the twin's practice notes attach to the fault mode, for example "span added loss, 0.5 to 15 dB", "amplifier gain change, 0.5 to 10 dB" or "WSS input port blocked". Each scenario maps to one change in the physics engine. For instrument faults the scenario is a bias of −10 to +10 dB or a frozen reading. The twin's practice notes add three scenarios that no vendor document gives: a channel-loading excursion on an OMS (−8 to +8 dB), a gain tilt across the band (−5 to +5 dB) and a slow loss of margin, simulated as a small receiver-side penalty (0.1 to 1 dB) because the twin has no ramp.

A failure of a port of a wavelength-selective switch (WSS) is simulated at the grain its documentation gives. That grain is the input port on which an OMS arrives at a ROADM, written "oms|port", and every channel arriving on that degree is blocked. The per-channel WSS modes are simulated at one channel, written "oms|slot". Three modes have no scenario: pump-current drift, laser-bias drift and the protection-switch hit.

What it compares. The twin compares summaries of the data in three windows around the frame's estimated onset:

- before: from 480 to 20 seconds before onset;
- during: from 20 seconds before onset to 120 seconds after;
- after: the last 300 seconds before the question, starting at least 150 seconds after onset.

The power and quality streams are amplifier input and output power, received power, channel-monitor power and OSNR, estimated OSNR, log bit error rate, DGD and SOP rate. For each, the twin takes the mean before, the mean after and the extreme during the onset. For the observed data it also takes their standard errors. It takes, in addition, the 15-minute counts of errored seconds of line ports. It compares nothing else: alarms, protection switches, client counts and health streams are not its evidence.

How it scores. For each candidate the twin simulates the scenario and computes a **misfit** in nats. The misfit is a negative log-likelihood: zero when prediction and observation agree and larger the worse they disagree. A difference is **standardised** by dividing it by its standard deviation. The misfit adds up the following items:

- for a power or quality stream, the **change** from before to after, observed against predicted. Its standard deviation combines the observed standard errors with the stated allowance for the twin's error on changes. It costs half the square of the standardised difference, at most 8 nats;
- the extreme during the onset (the lowest power, the highest bit error rate), compared in the same way, at most a further 8 nats (not for DGD);
- for the SOP rate, the peak during the onset on a logarithmic scale, with a standard deviation of 0.35 decade, counted only where observed and predicted peaks differ by more than half a decade;
- a signal lost in one and present in the other, 6 nats (3 nats when they differ only during the onset);
- each 15-minute count of line errored seconds, with an allowance of 2 plus 20 per cent of the larger count.

Comparing changes rather than levels removes most of the twin's input error and the instruments' calibration offsets. Both are common to the readings before and after onset, so they cancel in the change.

Fitting. The twin does not take a scenario's parameter, such as the size of an added loss, as given. It fits it, choosing the value that best explains the data. For each candidate it tries an eight-point grid over the documented range (logarithmic when the range is wide). It then refines the best point by six golden-section steps, a standard one-dimensional search that narrows the interval around the minimum by a fixed ratio at each step. For a transient it does this for every documented duration: polarisation transients of 5 ms to 5 s, mechanical disturbances of 0.2 to 60 s and channel-loading excursions of 0.02 to 3 s. A compound is scored with its causes' changes applied together, each cause at its own best value. A single fit costs about 20 simulations, a few seconds.

Credit only for explaining transmission. A hypothesis earns credit from the twin only if it explains the transmission data better than no fault at all. If its best fit improves on no fault by less than 0.5 nat, it receives the lower of its fit and the background, so it scores no better than a cause nobody proposed. Faults of equipment behaviour alone (an OTN switch failure, a cross-connect misconnection, a client port failure) change nothing the twin predicts and fall here, as do hypotheses fitted to almost nothing. The documentation terms judge them.

Causes with no scenario. A cause for which no scenario exists is fitted as predicting no change in transmission. Its twin score is then the lower of the no-fault score and the background, as for a fault of equipment behaviour alone. Its residual (Section 4.5) is every changed reading, and checks and the planner predict for it the answer with no fault. A compound containing such a cause keeps the fits of its other causes. This treatment corrects a design pitfall found and fixed, and the pitfall is instructive. A cause the twin could not simulate was once given the background value as a real score. Since the background fits every changed reading to about one standard deviation, the untestable cause then beat every testable cause that left readings unexplained, without any evidence for it. It also suppressed the misfit signal and gave the ablative step no residual to work from. A tool's ignorance must cost a hypothesis what an explanation of nothing costs, never what a good but unnamed explanation would score.

The background. The twin scores "none of these" as a correct but unproposed explanation would typically score. Such an explanation would leave unchanged what the data show unchanged, so those items are scored as the no-fault twin scores them. It would also fit every changed item to about one standard deviation, so each changed item costs half a nat. The background is then charged a further allowance of 12 nats, set on development cases. The size of the allowance is a balance. Too small an allowance lets "none of these" take the probability from a true cause: at 3 nats it takes 0.995 of the

probability on a development case of a soft fault. At 12 nats every true cause in the development cases scores above the background.

What it gives out. The twin's term gives the background and, for each candidate, the fitted log-likelihood with its fitted parameters.

Coverage. Fitting every candidate would cost minutes a case, so the twin is fitted only to the candidates that matter, as they come to matter. It is fitted first to the twelve candidates with the largest prior. Then, before every decision of the ledger, it is fitted to any of the six leaders and any contender of the planner (Section 4.10) that it has not yet scored. When "none of these" leads, the twin first scores every proposed cause it has not yet scored. Before the agent may conclude that no proposed cause explains the data, every proposed cause must have been tested, including one proposed with a small prior, such as a fault of the monitoring. The candidates of a return pass are fitted in the same way. The ledger cannot commit while the leader or the runner-up lacks the twin's term (Section 4.1), so every commitment rests on it. When the twin comes to cover a hypothesis, every check already made is read again for that hypothesis (Section 4.9), so each check speaks of every hypothesis for which the twin can predict it.

Predicting checks. The twin also predicts the answer each measurement check would give under each candidate. It performs the check in its own network with the candidate's fitted fault, noise-free, at the time the answer would be read. It then adds its calculation error, fixed per field and check target as above. Repairs made by the agent's earlier checks (a connector cleaned or reseated, a unit swapped) are applied first, as they were applied in the network.

To read each answer at the right time, the twin keeps a case clock. The clock starts at the question and moves to the time of each answer as checks complete; a check that proves unavailable does not move it. For an equipment check the twin returns no prediction ("not modelled"), and the check is read by its documented meaning (Section 4.9). The advisor plans from these predictions, and the interpreters score answers against them.

4.5 The twin used ablatively

A leader can be right and still not be the whole story: a second, quieter fault can hide behind a louder first one. The twin is therefore also used **ablatively**. The leader's fitted effect is subtracted from what was observed, and a second cause is fitted to what remains. This **ablative unmasking** of a coincident fault has two parts: a test of what the leader leaves unexplained and a search for the cause of it.

The residual. Once the leading hypothesis is fitted, the readings its fitted prediction leaves unexplained form its **residual**. A transmission item (a stream or a line errored-second count) is unexplained when the prediction misses it by more than 4.5 nats, which is three standard deviations. The documented propagation of the leader's causes acting together is also compared with the alarms and service effects in the frame, and those it does not produce are unexplained too. A fault of the monitoring raises no alarm and moves no service, so with an instrument fault as leader every observed alarm and service effect counts as unexplained.

The residual test. Before the ledger commits, the residual test asks whether the leader leaves too much unexplained to be the whole story. The commitment is deferred, and the loop returns to generation to look for a further cause, when one transmission reading lies more than 8 nats from the leader's prediction or when at least two items are unexplained. The threshold is low because a second fault behind a louder first one often leaves only one or two readings unexplained. Each leader is tested once.

The search. On the return, the agent searches where the residual points, anywhere in the network and not only within the frame's scope. It builds frames anchored at up to three of the residual's elements, with

observations restricted to the residual's instruments. On them it runs the backward walk and forward check of the abduction walk (Section 3.5). It pairs the leader with each of up to six causes proposed there, as compounds.

Fitting a pair. A pair is fitted in order. The leader's fitted changes are held, and the later cause is fitted given them. That fit includes the later cause's onset, tried at 0, 60, 120, 180 and 300 seconds after the leader's, since coexisting faults need not start together. A pair therefore stands only if its second cause explains what the leader leaves over, at a time the data allow.

Checks on a pair. A check read by its documented meaning (Section 4.9) speaks of one place, so a pair needs a rule for it. The check is read for the one cause of the pair that lies within its scope: the lightpath's signal path or the span, OMS, unit, switch, port or ODU path the check names. If no cause of the pair is in scope, the check is read as it is read for a cause elsewhere. If several are, it is not read for that pair.

Its limit. A second fault is invisible to the residual test when its transmission effect lies within the twin's error and it produces no alarm or service effect beyond the leader's.

4.6 Graph consistency and the documented equipment term

Two tools judge candidates from the vendors' documentation rather than by simulation. Both rest on the propagator's predictions of what a fault would cause (Section 3.4). Graph consistency tests each candidate against six rules. The documented equipment term counts the alarms and service effects a candidate fails to account for and so covers the equipment behaviour that the twin does not predict.

What graph consistency is. Six rules each ask whether a candidate is consistent with the graph and the frame. Each answers supported, contradicted or unresolved, with a reason. Where an absence could not be read from the records, the rule also records a request for the missing record. The rules are what an engineer would check on a whiteboard: "if it were this, we would also see that; do we?" Table 6 lists them.

Table 6. The graph-consistency rules

Rule	Applies to	Meaning	Verdicts
R1 Required effects	Each equipment cause with documented effects; a compound on its causes acting together	Every effect the documentation says the cause must produce, at an instrument that was examined, was seen	Contradicted if a required effect at an examined instrument is missing; unresolved if the missing effects fall at silent instruments; supported if all were seen
R2 Services	Each equipment cause, including one whose mode documents no effects for its element; a compound on its causes acting together	Every service the cause must take down is down, and every service down is one it could take down	Contradicted on either failure; supported if it accounts for all services down; unresolved if none is down

Rule	Applies to	Meaning	Verdicts
R3 Symptoms explained	Each equipment cause; a compound on its causes acting together	The cause explains at least 70 per cent of what was observed	Supported at 70 per cent; otherwise unresolved, since another cause may be at work
R4 Documented	Each equipment cause	The fault mode is documented for this equipment	Supported if documented; otherwise unresolved, since documentation is complete only as far as it goes
R5 Shared risk	Causes on two or more spans	The plant records place the spans in a common duct or cable	Supported if they do; otherwise unresolved, since shared-risk records are incomplete
R6 Monitoring	Instrument faults	An instrument at fault changes its own readings only; a real change would reach the instruments downstream on the same signal paths	Contradicted if readings downstream changed with it; supported if downstream instruments were examined and saw no change

Causes acting together. A compound, the cut of a shared duct included, is judged by R1, R2 and R3 on the documented propagation of its causes acting together, not cause by cause. The optical and digital effects of every cause accumulate before the services are judged. Protection is judged once, and where two causes act on one ODU the stronger effect stands. A working path hit by one cut and its protecting path hit by another therefore take the service down. A protection-unavailable alarm that one cut alone would raise is not required when the other cut takes the service down. Judged cause by cause, a true duct cut would contradict itself.

Completeness of the records. The rules read absence according to the declared completeness of each relation (Section 2.5): a missing record counts as absence only where the records declare the relation complete. Shared-risk records and documentation are declared incomplete, so R4 and R5 can never contradict. Only R1, R2 and R6 can.

How it scores. Each contradicting rule costs 4 nats, with at most two counted; support and unresolved verdicts cost nothing. Graph consistency reads both kinds of evidence in the data held, the changes in readings and the alarms and service states, at tau 0.3 on each. A contradiction therefore moves a candidate by 1.2 nats in the ledger. A candidate the rules contradict is also excluded from the generation of return passes.

Why R6 looks only downstream. R6 compares an instrument only with the instruments downstream of it on the same signal paths. Compared with readings anywhere in the network, it would contradict a true misreport whenever an unrelated reading elsewhere moved.

The documented equipment term. This term judges, from the documentation, the equipment behaviour that the transmission twin does not predict. For each hypothesis the propagator predicts the alarms and service effects (services down and services switched to protection) of its causes acting together. The term charges 2 nats for each alarm or service effect in the frame that the prediction does not produce. It also charges 2 nats for each one the prediction requires that was examined and not seen. The total charge is at most 8 nats.

Two hypotheses are charged in a particular way. A fault of the monitoring raises no alarm and moves no service, so every alarm and service effect in the frame counts against it. “None of these” is charged one item (2 nats).

The term takes the share of the equipment evidence that graph consistency leaves, tau 0.7. In effect it makes the comparison that an equipment-predicting twin would make, from what the vendor documents.

4.7 Health levels and margin levels

A slow trend has no onset in the window the framer examines, and the twin judges changes at an onset, so neither the change detector nor the twin sees it. Two terms read the present level instead. Each enters at tau 1.0 on its own evidence, which no other term reads.

Health levels. The health term is the evidence for the ageing of pumps and lasers. What speaks to such a drift is the present level of the health telemetry: a pump drive current or a laser bias current well above nominal. For each candidate at an element with health telemetry, the documented effects of its mode are matched to the health streams that see them. A mode documented only for another vendor’s equipment of the same class is matched through that documentation.

- If the mode accounts for a health stream, a level at or below 101.5 per cent of nominal contradicts it (4 nats), and a level at or above 103 per cent supports it.
- A candidate at an element with a high level that its mode does not account for is charged as the twin charges a reading it leaves unexplained: half the square of the level’s distance from the fleet’s median, at most 8 nats. The distance is measured in units of the fleet’s own robust scatter, 1.4826 times the median absolute deviation over every element reporting the stream, with a floor of 0.1 per cent. Most units are healthy, so the fleet sets what a normal level is.
- When some candidate accounts for a high level, “none of these” is charged 3 nats.

Candidates at elements without health telemetry are unaffected.

Margin levels. The margin term is the evidence for a slow loss of margin at a receiver. It compares each receiver’s pre-FEC bit error rate before the onset with the twin’s prediction of it with no fault, built from the records. The yardstick is the operator’s stated accuracy on predicted margin, 0.25 dB, which at half a decade of bit error rate per dB is 0.125 decade. A receiver whose rate is worse than predicted by more than three times that yardstick shows a loss of margin; one within one times it shows none. As a refinement after the freeze (amendment 24), the term can instead judge each receiver against the fleet: with `ADAGENT_MARGIN_FLEET=1` it subtracts the median departure of all receivers from the twin before applying the thresholds, so that a shift common to the twin does not read as lost margin everywhere. Off, the code behaves as frozen.

- An ageing mode whose documented effect the receiver’s bit error rate sees, whether documented for the element or borrowed from another vendor’s equipment of the same class, is supported where the receiver shows a loss and contradicted (4 nats) where it shows none.
- Any other hypothesis about that receiver or its instruments that does not account for a loss it shows is charged as the twin charges an unexplained reading: half the square of the deviation in units of the yardstick, at most 8 nats.
- When some candidate accounts for a loss, “none of these” is charged 3 nats.

Receivers with no loss and hypotheses elsewhere are not affected. The thresholds come from the operator's stated accuracy, not from the cases.

4.8 Checks and their effort

Checks are the tests the agent can order when the data held do not settle a case. Each costs effort, and some send a technician to a site, so choosing checks is where the agent spends its effort. Thirteen checks can be made, in two classes. **Remotely triggered tests** run from the operations centre. **Field tests** need a technician at a site, so they also incur travel to any site not yet visited in the case. Table 7 lists them; the minutes are the check's own duration, before any travel.

Table 7. The check catalogue

Check	Class	Minutes	Service impact	Returns
Embedded OTDR	Remote	8	None	Trace of the span: events by position, breaks and loss steps with size (where the upstream site has an embedded OTDR)
Performance detail	Remote	2	None	One-minute bins from 20 minutes before the question to the time of reading: minimum received power and errored seconds
Coherent diagnostics	Remote	2	None	Now: estimated OSNR, pre-FEC bit error rate, chromatic dispersion, DGD, SOP rate and margin; or signal lost
Transmit power	Remote	2	None	Transmit power or none if the laser or card failed
Cross-connect audit	Remote	15	None	Switch reachable or not; cross-connects differing from provisioning
Tandem connection monitoring	Remote	10	None	Status of each switch-to-switch segment of an ODU path
Loopback	Remote	10	Interrupts the service	Pass or fail at a client port
Spare-receiver retune	Remote	30	Interrupts the channel	Whether errors clear on the spare port
Portable OTDR	Field	30	None	As the embedded OTDR
Power meter	Field	20	None	Launch power per channel into an OMS, or none
Inspect and clean	Field	20	Brief hit	Dirty (cleaned), unseated (reseated) or clean; repairs what it finds
Span loss	Field	45	None	Measured and recorded loss of a span, from both ends
Unit swap	Field	60	Interrupts what the unit carries	Symptoms cleared or unchanged; repairs a unit fault

Seven of the checks measure transmission and are predicted by the twin: both OTDR traces, performance detail, coherent diagnostics, transmit power, the power meter and span loss. The other six are equipment checks, read by their documented meaning.

Travel and failure. Travel to a site not yet visited in the case takes 1 to 4 hours (median 2), drawn once per site per case. The agent does not know the draw and plans with 2 hours. Tests at a site already visited share the visit. Three per cent of remote tests fail, and a failed check is not evidence. The times are judgement from general operations practice, fixed in advance of every case. Effort is always reported with site visits beside it, since dispatch is the expensive decision in practice.

When a check is answered. The world answers a check when it completes: at the question time, plus the time of the checks made before it, the travel to any site not yet visited and the check's own minutes. A failed check takes its time as well; an unavailable one takes none. Each answer carries the time of its reading, from which the twin sets its clock. A check repeated in a case is a fresh trial, with fresh instrument noise and, for a remote test, a fresh chance of failure.

Two answers need a word on their content. The performance-detail register holds one-minute bins from 20 minutes before the question to the time of reading, so the onset stays in view however late the check is made. Coherent diagnostics report the estimated OSNR, pre-FEC bit error rate, chromatic dispersion, DGD, SOP rate and margin, each with the instrument noise of Table 3.

4.9 Check interpreters and the accuracy standard

When a check is answered, an interpreter turns the answer into a likelihood term for the ledger. Every interpreter rests on one rule for deciding whether a difference between two answers or between an answer and a prediction, is larger than the combined error of the instruments and the twin. That rule is the **accuracy standard**. The agent and the harness use the same standard, so the agent is scored by the rule it reasons with.

The accuracy standard. Each numeric field of an answer is given a standard deviation that combines four sources of error:

$$\sigma = \sqrt{(\text{instrument}^2 + \text{twin}^2 + (\text{rel} \times |\text{predicted}|)^2 + (\text{change} \times |\text{predicted} - \text{no-fault}|)^2)}$$

- *instrument* is the instrument's accuracy, calibration included;
- *twin* is the twin's absolute accuracy on that quantity, input and calculation error together;
- *rel* is an error proportional to the value;
- *change* is the twin's error on a predicted change, as a fraction of the difference between the predicted answer and the answer predicted with no fault.

Table 8 gives the values, field by field, as the practice notes state them. Bit error rate and SOP rate are compared on a logarithmic scale.

Table 8. The accuracy standard, field by field (one standard deviation)

Field	Unit	Instrument	Twin	Proportional	Fraction of a predicted change
Minimum received power (performance detail)	dB	0.31	0.2	0	0.15
Transmit power	dB	0.31	0.2	0	0

Field	Unit	Instrument	Twin	Proportional	Fraction of a predicted change
Power per channel at an OMS input (power meter)	dB	0.1	0.2	0	0.15
Estimated OSNR	dB	0.42	0.55	0	0.2
Margin	dB	0.2	0.25	0	0.2
DGD	ps	0.3	0.3	0	0.2
Pre-FEC bit error rate	decade	0.1	0.2	0	0.2
SOP rate	decade	0.04	0.15	0	0
Span loss, OTDR loss step	dB	0.1	0.65	0	0.15
Errored seconds	s	0	2.0	0.2	0
Chromatic dispersion	ps/nm	1.0	50	0.05	0

How a difference is costed. A numeric field costs half the square of its standardised difference, at most 8 nats. A categorical difference costs 5 nats: lost against present, a different set of OTDR events or a different finding. An answer's misfit is that of its worst field. The fields are not added, because the fields of one answer share their errors: margin, OSNR and bit error rate follow one signal-to-noise ratio, so adding them would count one error many times. Two answers are **alike** when the misfit between them is below 2 nats, that is, when every field lies within two standard deviations.

A series of readings. Performance detail returns a series of one-minute readings of one receiver, often twenty or more. Its errors arise in two ways. The receiver's calibration offset and the twin's own error are the same in every reading, while the reading noise varies from one reading to the next. The series is compared accordingly. The offset common to all the readings (the median difference between answer and prediction) is compared once, against the calibration and twin accuracies. Each reading's departure from that common offset is compared against what varies by reading: the reading noise (0.1 dB for received power), the error proportional to the value and the twin's error on a predicted change. The worst of the departures is judged as a single reading equally rare would be judged, by mapping its probability among n readings to one reading. This avoids a pitfall of any standard that judges an answer by its worst element. Judging a long series by its worst reading against the tolerance of a single reading would penalise a correct prediction for the length of the series: at the test bed's error sizes it charges a correct prediction more than 2 nats in about 12 per cent of series, where the comparison above does so in about 8 per cent, close to the rate intended for each of its two sources of error.

Two ways of reading an answer. A check's answer is scored for each hypothesis in one of two ways, depending on whether the twin can predict it.

Against the twin's prediction. When the twin covers a hypothesis and the check measures transmission, the answer is scored by its misfit against the twin's prediction under that hypothesis, at the time the answer was read. The no-fault prediction at the same time sets the size of the predicted change.

By the check's documented meaning. For an equipment check and for a single cause the twin does not cover, the answer is read as the check catalogue documents it. A contradiction costs 4 nats and support costs nothing. The documented meanings are these.

- A **unit swap** that clears the symptoms supports a fault of that unit itself and contradicts contenders elsewhere. The faults a swap repairs are amplifier, transmitter, card, receiver and client-optic faults and the ageing of a pump or laser. A swap that leaves the symptoms unchanged contradicts a fault of the unit and says nothing of contenders elsewhere.
- **Inspection** that finds a dirty or unseated connector supports a connector fault at that OMS input and contradicts contenders elsewhere. Finding the connectors clean contradicts the connector fault.
- A **loopback** that fails supports a failure of that client port and contradicts contenders elsewhere. One that passes contradicts the port failure.
- A **cross-connect audit** that finds the switch unreachable supports a failure of that switch. One that lists a path's cross-connect as differing from provisioning supports a misconnection of that path at that switch. Either finding contradicts contenders elsewhere. An audit with no finding contradicts both faults at the switch; an unreachable switch says nothing of a misconnection at it.
- A **spare-receiver retune** whose errors clear on the spare port supports a receive-side fault of that receiver (receiver degradation or card failure). It contradicts every line contender and receive-side faults elsewhere. Errors that persist contradict the receive-side fault at that receiver and are consistent with every line contender. A slow loss of margin is documented as ageing along the path, not as a fault of the receiver, so the retune does not read it as receive-side.
- **Tandem connection monitoring** reports the status of each switch-to-switch segment of an ODU path, and it is read from which segment the documentation says a contender would fail. A segment fails when the signal into its far-end switch is lost in the segment's own direction or when that switch (or its cross-connect for the path) fails. So a switch failure or misconnection at a segment's far-end switch is expected to fail that segment. So is a loss of signal on the segment's lightpath in its direction (fibre cut, pulled connector, pump failure, transmitter or card failure, WSS port failure); a transmitter fault at the receiving end does not fail it. The answer supports a contender whose expected segment failed and contradicts one whose expected segment did not. Any failed segment contradicts contenders expected to fail none. The check says nothing of a soft fault on the path, which may or may not fail a segment, or of a misconnection placed on a switch with no path named, which may or may not lie on this path.

What enters the ledger. A hypothesis about which neither way says anything is declined, as is a compound the twin does not cover. Every check term also gives a value to "none of these", the chance that the cause is one nobody has proposed. On a check the twin predicts, an unproposed cause is charged as fitting the answer to within two standard deviations (2 nats). An answer the leading candidate predicted therefore lowers "none of these", and an answer no candidate predicted raises it. On a check read by its documented meaning, an unproposed cause is read as lying elsewhere: a finding at the place checked counts against it and a negative finding does not.

This value is a design choice with a measurable trade-off. Without it, a check that confirms the leader cannot lower "none of these", and the agent abstains in cases that a confirming check has in fact settled. With it, a correct prediction that happens to miss by more than two standard deviations on one check passes some of its probability to "none of these"; that happens when a calibration offset and a twin error add up, as occasionally they will. Which cases gain and which lose was not measured separately.

The verdict recorded for a hypothesis is contradicted when the answer costs it 3 nats or more and supported when it costs 1 nat or less. A check that failed or was unavailable gives no term. Each check is its

own evidence at tau 1.0. Every earlier check is applied to hypotheses that arrive later and read again for hypotheses the twin comes to cover.

4.10 The advisor

The advisor is the planner of the agent as reported: deterministic code that chooses the check expected to reveal most per hour of effort. It works in four steps. It picks the contenders, lists the checks that bear on them, predicts each check's answer under each contender and scores each check by the information it is expected to give. A fifth step, "can this decide?", screens out costly checks that could not lead to a decision.

Contenders. The advisor's **contenders** are the named members of the set the ledger would report now (the leaders holding 90 per cent of the probability, Section 4.2), at most twelve. The set is filled by probability to at least three named hypotheses, so that a leader short of commitment can still be confirmed against its runners-up. The planner therefore works on the same set by which a stop is judged (Section 5.3).

Checks and their effort. The advisor lists every check of the catalogue that looks at the contenders' locations, with its effort: its minutes, plus 2 hours for each new site, plus a 1-hour penalty for interrupting service. A check whose answer stands is not repeated; one that failed may be tried once more.

Predicted answers. For each check the advisor predicts the answer under each contender at the estimated time of reading, which is the case clock plus the estimated travel and the check's minutes. Where the twin covers the contender and the check measures transmission, the twin predicts the answer. Otherwise the check's documented meaning predicts it, from the answers the catalogue documents for the check. The advisor takes the answer that the contender's reading fits (misfit zero). Failing that, it takes an answer the reading says nothing against. Failing that too, it takes the answer the reading fits best, where the answers differ in fit. Where neither way gives a prediction, the contender's answer is unknown.

Expected information. The advisor groups the contenders whose predicted answers are alike by the accuracy standard (Section 4.9), with the no-fault prediction setting the size of a predicted change. A check can separate contenders only if it puts them in different groups. "None of these" joins as a group of its own, with its probability, in two cases: when the leader's predicted answer differs from the no-fault answer, or, for a check read by its documented meaning, when the leader's answer is a finding that would count against a cause anywhere else. Such a check can confirm the leader against a cause nobody has proposed, because the ledger lowers "none of these" when the answer comes back as the leader predicted (Section 4.9). A check that would read normal whatever the cause confirms nothing and gains nothing from "none of these". A check's **expected information** is the entropy, in bits, of the contenders' probability over the groups, scaled by the share of probability whose answer is known.

Unknown answers count as unknown. A contender about which a check can say nothing counts as unknown, exactly as the interpreters treat it. Counted as giving a definite answer, it would make spare-receiver retunes and inspections look informative when they cannot move the posterior.

"Can this decide?" Before recommending a check whose effort is half an hour or more (a site visit or a long test), the advisor asks whether the check could lead to a decision. From the answer groups it has already formed, it computes for each possible answer whether the ledger would then commit, either at once or after one more check of the same strength. From its present beliefs it then computes the probability of an answer that would do so. If that probability is under 5 per cent, the check is refused and the advisor considers the next-best check. Remote reads of a few minutes are never refused, because their effect on belief is often larger than the agent can foresee. The half-hour condition and the 5 per cent threshold were fixed before the step was run on any case.

The choice. The advisor recommends the admissible check with the most expected information per hour, provided it reaches 0.05 bit and fits the remaining budget. When no check qualifies or fewer than two contenders remain, it stops, and the agent reports its set with the reason. The advisor never asks to return to generation; in the agent as reported, returns come from the ledger's signals and the residual test.

4.11 The LLM orchestrator

The orchestrator is the alternative planner, measured in the paper: an LLM chooses the next action. The two planners differ only in who chooses; in both, the ledger alone holds the probabilities and decides commitment. The orchestrator never writes to the ledger, and it never sees the truth or the advisor's information scores.

What it is shown. At each step the orchestrator sees:

- the question, the effort left and the sites visited;
- the leading candidates (the advisor's contenders) with their probabilities and the probability of "none of these";
- each generation pass with its trigger and the number of new candidates it produced;
- the returns left;
- graph consistency's verdicts on the leaders and the twin's fit of each leader;
- the checks made, with their answers;
- up to 25 checks available at the leaders' locations within the effort left, each with its class, effort and service impact.

Beside each check the candidates are **grouped by the answer it would give**. The groups are the advisor's own, formed from the same predictions and by the same accuracy standard, and they include the "none of these" group with its probability and its meaning. A check is marked "separates nothing" only when the advisor would find no group to separate. The "can this decide?" step is applied, as the advisor applies it, to every check before it is offered: a check the step refuses is not shown, and the state says how many were withheld and why.

What it is told. The instructions cover three things.

- The actions: make a listed check; return to generation, which tests nothing, is limited and is no substitute for a check that separates candidates; or stop, saying why.
- The tolerances, in round figures, within which answers count as the same.
- How the outcome is judged. A commitment is judged against the truth. Stopping is judged right only when no check within the effort left could separate any candidate in contention, including candidates of low probability, dispatches and checks that interrupt service.

It is also told to prefer the cheaper of two checks that separate the same groups and otherwise the one that moves the most probability per hour.

What it returns. It returns an action, a check index where the action is a check and a one-sentence reason, as structured output. A failed call or an invalid index becomes a stop. The fixed instructions come first so that, after the first call, they are billed at the cached rate. The LLM is gpt-5.6 (Sol), and the prompt is fixed and recorded with its version in the code.

Design pitfalls. An LLM planner can fail for reasons that lie in the architecture around it rather than in the LLM's judgement. Five such pitfalls shaped the orchestrator, and they apply to any architecture that puts an LLM in a planning role.

- **Raw predictions.** An LLM shown, for each check, the raw answer the twin predicts under each candidate reads differences far inside the twin's error as real and chooses checks that cannot separate anything. Hence the grouped answers and the stated tolerances.
- **An action the loop cannot grant, turned silently into another.** A loop that treats a request for a return to generation, made after all the allowed returns are used, as a request to stop ends the case without the planner having chosen to end it. The loop instead refuses the return, says so and asks the planner again.
- **A state that leaves out what the planner needs.** A planner not told how many returns are left or that earlier returns produced no new candidates, asks for yet another return. Hence the record of passes in the state.
- **A criterion that differs from the one it is judged by.** A planner told that stopping is right when no check could separate "the leading candidates", while the outcome is judged over every candidate reported, declines checks that could still narrow the set. Hence the stopping rule stated as the outcome is judged.
- **Different inputs from the coded planner.** A planner shown groups over the named candidates only cannot see that a check whose candidates all predict the same answer can still confirm them against a cause nobody proposed. It then stops where the advisor would make a two-minute check and commit. Hence the advisor's own groups, "none of these" included.

The lesson is that a planner is only as good as the state it is shown and the criteria it is given. Both must be the ones the system actually applies, and an action the system cannot grant must be refused visibly, never converted into another.

4.12 The loop

The loop is the control flow of a diagnosis. It runs generation, the tools, the planner and the ledger in a fixed order.

1. **Generation, first pass.** The six general lines, the three specialised lines and the transient event recogniser propose candidates (Section 3).

2. **The terms on the data held.** Graph consistency, the documented equipment term, the health levels and the margin levels judge every candidate. The twin is fitted to the twelve with the largest prior.

3. **Coverage before each decision.** Before every decision, the twin is fitted to any of the six leaders and the planner's contenders that it has not yet scored and to every proposed cause when "none of these" leads (Section 4.4).

4. **The decision step, repeated.** The loop then repeats the decision step described below until the case ends.

5. **Limits.** At most three returns to generation in all, whatever triggered them; at most twelve checks; at most 8 hours of effort. A case ends in **non-convergence** when it reaches the check or effort limit or when its planner stops with a quarter of an hour or less left.

6. **The end of the case.** A final snapshot of the ledger is recorded before the decision, so that the recorded trajectory ends with the posterior on which the case is scored. When the decision is a set, the commitment at a coarser grain is then sought (Section 4.2).

The decision step. At each turn the first of the following that applies decides what happens.

- If the ledger would commit, the residual test (Section 4.5) is run on the leader. If the test defers the commitment, the loop returns to generation. Otherwise the ledger commits and the case ends.
- If the ledger signals regenerate or misfit, the loop returns to generation.
- Otherwise the planner is asked for the next action. A stop ends the case. For a check, the check is made and its effort counted, the twin's clock is moved to the time of its answer and any repair it made is recorded. Its answer is interpreted and added to the ledger, and the hypotheses it contradicts are excluded from later generation. A return asked for by the orchestrator when none is left is refused; the orchestrator is told so and asked again.

Return passes. Each return to generation works from a new frame, built for the question with a widening factor that doubles per pass (2, 4 and 8). Generation runs on that frame, and graph consistency and the documented equipment term are rebuilt on it and judge every hypothesis again, old and new, so that all are judged on the same evidence (Section 3.15).

4.13 What the twin must predict

The twin is the agent's most demanding tool, and an operator deciding whether its own twin can serve this architecture needs to know exactly what is asked of it. Five predictive qualities are used.

1. **The transmission state under a hypothesised fault, with the fault's parameter fitted.** Given a documented scenario (a span's added loss, an amplifier's gain drop, a receiver penalty, a blocked WSS port, an instrument's bias), the twin must predict the transmission state at every deployed instrument. That means the levels of power, OSNR, bit error rate, DGD and SOP rate before, during and after the onset, as well as whether the signal is lost. It must do so fast enough to fit the parameter by about 20 simulations per hypothesis, for a dozen or more hypotheses a case.

2. **Absolute accuracy, measured and stated.** A check's answer is compared with the twin's predicted value, not with a change, so the twin's error on levels must be known and stated field by field. The statement here (Table 8) is 0.25 dB on margin, 0.55 dB on OSNR, 0.2 dB on power, 0.65 dB on span loss, 0.3 ps on DGD and 0.2 decade on the bit error rate. The test bed's twin meets it, at 0.24 dB on margin measured over the network. The same stated accuracy on margin is the yardstick of the margin levels (Section 4.7). The paper's robustness test measures what happens when the twin's predictions are shifted by about twice that statement: a level read against the twin carries the shift in full, and the margin term finds 166 of 365 receivers short of margin, unless it judges each receiver against the fleet.

3. **Accuracy on changes.** The data already held are compared as changes from before to after onset, so the twin must state its error on a predicted change: a floor plus a fraction of the change (Section 4.4). The test bed's twin errs by 5 per cent of a change.

4. **The answers of measurement checks.** For coherent diagnostics, performance detail, transmit power, the power meter, span loss and OTDR traces, the twin must predict the answer at the time of reading, after the repairs of earlier checks.

5. **Noise-free predictions with its own error.** The twin must predict what the instrument would read without noise, with an error of its own that is fixed for the case and independent of the world's noise. The

accuracy standard then adds the instrument's accuracy and counts each answer at its worst field, because the twin's error and a calibration offset repeat across the fields and bins of one answer.

What the twin is never asked. The twin is never asked to predict equipment behaviour: the alarms raised, their soak and suppression, protection switching, client-side counts or the answers of equipment checks. Each of those is covered from the documentation.

- Graph consistency tests the documented required effects, services and symptoms of each candidate (R1 to R3), with compounds and duct cuts judged on their causes acting together.
- The documented equipment term compares the alarms and service effects in the frame with the documented propagation of each hypothesis's causes acting together.
- The equipment checks are read and planned by their documented meaning (Section 4.9).
- In the residual test, the alarms and service effects that the leader's causes do not produce come from the documented joint propagation.

A twin that also predicted equipment behaviour would let these comparisons be made by simulation instead of documentation. Where the documentation's predicted alarms and service effects disagree with how the equipment behaves, such a twin would judge better; where the documentation is silent, as it is on the protection-switch hit, only such a twin could judge at all.

5. Scoring and costing

5.1 The cases

A **case** poses one question about one anomaly at a point in the history. The question is what an operator would ask. Three examples (svc024 names a service; trx.och185.z and trx.och177.z name transceivers; times are seconds into the simulated case):

- "The customer on service svc024 (100GE between S05 and S08) reports a brief interruption from about t=660 s. What is the cause?"
- "Monitoring shows the received power reported by trx.och185.z unchanged since about t=600 s while neighbouring readings vary. What is the cause?"
- "Monitoring shows the transmit laser bias current of trx.och177.z rising over about 14 days, with no alarm and no service affected yet. What is happening, and is action needed?"

In the diagnosis cases the fault begins at t = 600 s and the question is asked at t = 1,200 s. The simulation stops at the question, so no data from after it exist.

Each case is generated from a numbered random seed. These are the sets used.

- **Development cases** (seeds 9100 to 9115, among others) fixed the settings of the agent.
- **Generation cases** are 200 cases from seeds 1000 to 1099, each posed at day 730 and at day 2190. Their faults are drawn by fault family and redrawn until the fault is noticeable enough that someone would ask about it, so the set is weighted towards hard failures. They measure the generation lines on their own.
- **Diagnosis cases** are 320 cases from seeds 2000 to 2319, 40 of each of the eight kinds of anomaly, so that every kind can be reported on its own. They serve the analyses that compare configurations of the agent on the same material; some studies use the first 160 (seeds 2000 to 2159, 20 of each kind).

- **Sealed cases** are 320 cases from seeds 7000 to 7319, 40 of each kind, generated as the diagnosis cases are. They were reserved before the final development of the agent, refused by the harness until the code was frozen and opened once, for the paper's headline results.
- **Completion test cases** are 160 cases from seeds 3160 to 3319, generated as the diagnosis cases are and used once, for the test of documentation completion.

Table 9 describes the eight kinds of anomaly.

Table 9. The eight kinds of anomaly

Kind	What happens	How it comes to notice
Hard	A loss of signal: fibre cut, pulled connector, amplifier pump failure, transmitter or card failure, WSS port failure, OTN switch or client port failure, cross-connect misconnection	Alarms and lost services
Soft	A degradation: added span loss, dirty connector, gain drop, gain tilt, noise-figure rise, DGD rise, WSS attenuation drift or filter shift, receiver degradation	Changed readings and errors, sometimes threshold alarms
Transient	A short event that has passed: SOP transient, mechanical disturbance of a span, channel-loading transient, protection-switch hit	Errored seconds and excursions; little or nothing lasting
Drift	Slow ageing: pump current drift, laser bias drift, margin erosion	A trend question, posed before any alarm or service effect
Monitoring	A fault of the monitoring: an instrument that misreports by a fixed bias or telemetry that has frozen	A question about a changed (or unchanged) reading
Two faults	Two independent faults a few minutes apart	As for each fault
Maintenance	A maintenance window at a site shortly before the fault; half the time the window caused it	As for the fault, with a recent change on record
Duct cut	A cut of a shared duct, taking down every span in it, whether or not the duct is on record	Many alarms and protection switches at once

5.2 Equivalence classes

Every case is scored by the **harness**, the part of the test bed that runs each case and judges its outcome. The harness works outside the agent and has access to the world simulator: the true network, the true losses and the true deployment of instruments. The agent's tools never see any of these. The scoring rules matter as much as the agent, because a diagnosis can only fairly be judged against what the data could have shown.

Some faults cannot be told apart on the data. A filter shift in a WSS and a receiver degradation both raise one receiver's errors while its power and OSNR stay normal. A rise in DGD on any span of a path changes only the DGD measured at the receivers of that path. Scoring a diagnosis against the truth alone would punish an agent for not knowing what nothing could show. For each case the harness therefore computes the truth's **equivalence class**: the causes that the deployed instruments could not tell apart from the truth. An alternative cause is considered at every location named by any hypothesis the agent held. It

belongs to the class if some value of its parameter (the size of an added loss, for example) makes it indistinguishable from the truth on everything the deployed instruments could show.

How membership is tested. Membership is tested by simulation. The harness simulates, without noise, every interval from the start of the case to the question: once under the truth and once under each alternative placed at the truth's onset. From each simulation it builds a **signature** of what the instruments would show. The signature holds the alarms raised, the protection events and the services down and receivers errored in each interval. It also holds every reading the deployed instruments take: amplifier powers, pump current, laser bias, received power, OSNR, log bit error rate, DGD and SOP rate, together with channel-monitor power and OSNR. Misreporting and frozen instruments are applied, so the readings are those the instruments would actually report. Two signatures differ outright if anything discrete differs or if a reading is lost in one and not the other. Otherwise their difference is the largest difference of any reading, measured in units of that reading's combined noise. The alternative's parameter is fitted to make this difference as small as possible, by a search over twelve points of its documented range refined by twelve golden-section steps. The alternative is a member of the class if its smallest difference is at most two standard deviations.

Naming. The agent names a WSS input port by the optical multiplex section (OMS) that arrives on it, written "oms|port". In the world the same fault is the failure of the WSS port on which that OMS arrives, and the harness treats the two as one cause.

Several parts. Cases with more than one fault or one fault with several parts, have a class for each part. For a case of two faults, the class has one **component** per fault: the alternatives that are indistinguishable when that fault alone is replaced and the other kept. A duct cut is one fault but cuts several cable sections, so its class has one **part** per section. Each part holds either fibre of its section and any cause indistinguishable from cutting that section while the rest of the cut is held. Below, both the faults of a two-fault case and the cable sections of a duct cut are called the **parts of the truth**; a single fault is one part.

5.3 Outcomes

Every case ends in one outcome. The agent either commits to a hypothesis, commits at a coarser grain or says "I don't know", and a case can also run out of checks or effort.

A **commitment** is scored in one of three ways.

- **Right**: the committed hypothesis names, for every part of the truth, a member of that part's class. A single cause indistinguishable from the whole of a duct cut is also right, since the data allow no better.
- **Partial**: it names members for some parts of the truth and misses others, for example one of two faults or one cable section of a duct cut.
- **Wrong**: any other commitment.

A right commitment is further marked **right where the data were ambiguous** when the agent's own checks left some other member of the class unseparated from the truth. The agent was right, but the evidence it gathered could not have told the truth from that member. To test this, the harness performs each check the agent completed again, without noise, at the time it was made and after the repairs of the agent's earlier checks. It does so once in a world with the truth and once in a world with the member at its best fit. The member is unseparated when the two answers are alike by the accuracy standard (Section 4.9).

A **commitment at a coarser grain** (Section 4.2) is scored by what it names.

- A single cause shared by the candidates is scored as a commitment to that one cause: right if it is the true fault, partial if it is one of several true faults, wrong otherwise.
- An element, an OMS, a site or a fault mode is scored **right at a coarser grain** if a true fault lies at that place or is of that mode, and **wrong** otherwise. It never counts as right.

The “I don’t know” outcome that a coarser commitment replaces is kept beside it, so that both can be reported.

An **”I don’t know”** is the agent’s report of the set of hypotheses still in contention (Section 4.2). The harness judges it either **justified**, meaning that nothing within the budget could have done better, or **unjustified**. The judgement depends on whether the set contains a right member, one in the truth’s class.

- If no member of the reported set is right, the “I don’t know” is justified only when the agent flagged the cause as probably not considered and the truth was indeed absent from every hypothesis the agent held. Otherwise it is unjustified.
- If a member is right, the harness asks whether some other member could have been separated from the truth. It considers the members a planner can act on: the leaders of the reported set, up to the planner’s cap of twelve (Section 4.10). A member is separable by the data already held when no value of its parameter makes it match the truth. Otherwise the harness tries every check in the agent’s own catalogue (Table 7) at the member’s and the truth’s locations whose minutes, plus the actual travel to sites not yet visited, fit the effort left. Each check is performed without noise in three worlds: with the truth, with the member at its best fit in place of the truth and with no fault. It is performed at the time it would complete (the agent’s stop, plus the check’s own travel and minutes) and after the repairs made by the agent’s checks. The check separates the pair when its answers under the truth and under the member are not alike by the accuracy standard; the no-fault answer sets the size of a predicted change. If any member is separable, the “I don’t know” is unjustified. If none is, it is justified.

The harness and the agent therefore judge what a check can resolve by one standard. A check that the agent’s tools would read as giving the same answer under two causes is not counted against the agent as one that separates them. A difference within the instruments’ noise or the twin’s stated accuracy separates nothing.

Two further outcomes complete the list. A case that reaches the limit on checks or effort (Section 4.12) ends in **non-convergence**. Manual troubleshooting has one outcome of its own: **no procedure applies**, when the case offers no alarm or degraded receiver for a procedure to start from.

Cause and place, separately. “Right” means cause and place together, but the two can come apart. The records also hold the agent’s final probability on the true kind of fault, wherever it placed it, on the true place, whatever kind of fault it named there, and on both.

5.4 What the judgement of "I don't know" means

The rule is deliberately strict on the agent. A justified “I don’t know” says that the agent narrowed the causes as far as anything within the budget could. An unjustified one says one of two things: either the truth was missing and the agent did not say so, or some check could have narrowed the set further. Such a check counts even if it would have ruled out only a low-probability contender and even if it interrupts service or needs a dispatch. The standard for a justified abstention thus comes from what the network’s own evidence could identify, not from the agent’s own reasoning.

5.5 Effort as scored

The study measures diagnosis only. Nothing that follows a diagnosis is counted for either the agent or manual troubleshooting: no repair, no restoration and no follow-up visit.

Effort is the time diagnosis spends on checks and on travel, in hours. It has two parts.

- **The minutes of every check requested**, at the fixed time of each check (Table 7). Remote tests are run from the operations centre and take minutes with no one leaving it: two minutes to read a receiver's performance detail, thirty to retune a channel to a spare receiver. Field tests need a technician at a site. A failed remote test still counts its minutes; a check that cannot be made counts nothing.
- **Travel** to each site visited for the first time in a case: one to four hours, drawn once per site for each case, so that the same site in the same case costs the same whoever is sent. Further tests at a site already visited share the trip.

Effort is split into **field hours**, the tests on site and the travel to them, and **remote hours**, the tests run from the operations centre. A **site visit** is the first trip to a site in a case. Visits are reported beside effort because sending a technician is the expensive decision.

Three things are deliberately not counted. Reading the data the network already holds (its alarms, counters and streams) costs nothing, for the agent and for the procedures alike. The engineer's own time, to read alarms, choose a procedure and interpret results, is not counted. The agent's computing time, the language-model calls and the tools' running time, is recorded separately as running time and is not part of effort.

Two field checks change the network as part of the test: swapping a unit to see whether the symptoms clear and inspecting and cleaning a connector to see whether a reading recovers. They are counted as diagnostic tests, by substitution and observation, which is how troubleshooting uses them. Whatever they repair is incidental and is not counted as repair.

Effort is reported by outcome. A total of effort over all cases adds hours spent on cases one approach diagnosed to hours spent on cases it ended without a diagnosis, and it counts at zero the cases the other approach never attempted. Two approaches can then show similar totals while one buys right diagnoses cheaply and the other buys almost nothing. The results therefore give effort and visits by outcome (right, right at a coarser grain, partial, wrong, "I don't know" and no procedure), paired on the cases both approaches got right and per right diagnosis. They show the effort spent on cases that end unresolved apart from the rest.

The agent and the procedures pay exactly the same prices, through the same check interface and the same costing in the harness. The comparison therefore rests on both paying the same prices, not on the prices being exact. **Service-affecting checks** are counted as the loopbacks, spare-receiver retunes, unit swaps and inspections that completed.

6. Every modelled quantity, its source and its test

Every result rests on modelled quantities. This section lists each quantity the results depend on, with its value, its source and how it was tested. A source is one of four kinds:

- a **standard** (ITU-T, IETF);
- a **specification** written for the test bed in the form of vendor documentation;

- **operations practice**, as written in the test bed's practice notes (`knowledge/practice/`);
- a **stated assumption**.

The test column names the arm or record that tests the quantity, or says that it was not tested. Every value was checked against the code frozen under manifest C4H.

6.1 Effort

Effort is the elapsed hours of the checks an approach commissions, remote and in the field, plus travel to each site a technician visits (Section 5.5). It is the same for the agent and for manual troubleshooting, and it is not person-hours.

Table 10. Effort

Quantity	Value	Source	Test
Duration of each check	2 to 60 minutes by check (Table 7)	Operations practice (`checks.yaml`): typical durations	Not tested
Travel to a site	1 to 4 hours, median 2, drawn once per site and case	Stated assumption, from operations practice	Not tested
Budget per case	8 hours of effort, at most 12 checks	Stated assumption	Not tested
Remote and field	Remote checks run from the operations centre with nobody on site; field checks need a technician	Operations practice	Reported separately in every table of effort

6.2 The network

Table 11. The network

Quantity	Value	Source	Test
Sites, hubs, links	24 sites, 8 hubs with OTN switches, 41 links of 175 to 647 km on a 2,000 by 1,000 km plane	Stated assumption: a seeded design typical of a national backbone	The specialists were trained on generated networks of 8 to 16 sites
Spans	336 spans of 61 to 99 km, 12 per cent aerial	Stated assumption, typical practice	Not tested
Shared risk	8 shared ducts and 2 shared cables; 5 of the 10 missing from the plant records	Stated assumption	Records degraded (`kgx_c4`, arm kg-inst)
Lightpaths	225 coherent lightpaths: 400G DP-16QAM up to 800 km, 200G DP-QPSK beyond; 64 slots of 75 GHz	Stated assumption, current coherent practice	Not tested

Quantity	Value	Source	Test
Services	400 as built, 116 with ODU SNCP and 4 with OCh 1+1; on day 2190, 380 in service (233 10GE, 137 100GE, 10 400GE; 107 with SNCP, 4 with OCh 1+1)	Stated assumption	Not tested

6.3 The knowledge graph

Table 12. The knowledge graph

Quantity	Value	Source	Test
What is recorded	Written only by lifecycle processes: onboarding, provisioning, ticket closure	Architecture principle	Knowledge graph degraded (`kgx_c4`)
Completeness by relation	Inventory and provisioning complete once recorded; ducts, cables and instrument registrations incomplete; documentation as written	Operations practice (`records.yaml`)	Records degraded (`kgx_c4`, kg-inst)
Schema	Layered as in the ITU-T functional model (G.805, G.798, G.874)	Standards	Relations flattened (`kgx_c4`, kg-flat)
Size on day 2190	7,718 nodes and 21,880 edges, 3,643 of the nodes instruments	Generated	Not applicable

6.4 Fault modes, rates and alarms

Table 13. Fault modes, rates and alarms

Quantity	Value	Source	Test
Fault modes	31 in the world, in six groups; 23 documented as faults; 26 vendor equipment types and two plant types	Specifications in vendor form; plant modes from operations practice	Missing documentation (`kgx_c4`, kg-sig and kg-modes; `a20`)
Documented rates	For example, fibre cut 1 per 1,000 route-km a year buried and three times that aerial; span added loss 0.03 per span a year	Operations practice (`fibre_plant.yaml`) and specifications	Not tested directly; the prior orders the search and the evidence decides (Section 4.1)

Quantity	Value	Source	Test
Case mix	Eight kinds in equal number, 40 of each in 320 cases	Stated assumption of the evaluation, not field frequencies	Reweighted over 20,000 random mixes (`mix_c4.json`): the agent stays ahead of manual troubleshooting in every mix
Alarms and suppression	G.798 defects and their suppression; vendor-form alarm definitions and probable causes	Standards and specifications	Missing signatures (`kgx_c4`, kg-sig)

6.5 Instruments

Table 14. Instruments

Quantity	Value	Source	Test
Noise, one standard deviation	Table 3	Stated assumption, typical datasheet figures	Not tested
Misreport	A fixed bias, drawn per case	Stated assumption	The cross-check capability (`c4_dev`, A-L6)
Freeze	The last value repeated on schedule	Stated assumption	Measurement liveness (`c4_dev`, A-L1)
Instrumentation tiers	Status quo: 18 sites at the coherent tier, 6 at the full tier; future: every site at the full tier	Stated assumption from the history's roll-out	Both branches reported (`c4`, arms with and without `-fut`)

6.6 The twin

Table 15. The twin

Quantity	Value	Source	Test
The gap	Fibre attenuation 0.20 dB/km (standard deviation 0.008), connector loss 1.0 dB (0.25), noise figure 5.5 dB (0.3), gain set-point error 0.3 dB, calibration offsets within ±0.5 dB	Stated assumption, centred on datasheet values	Measured error on 450 receivers (`c4h_cloud/twin_fidelity.txt`, `twin_error_2026-11-04.txt`)

Quantity	Value	Source	Test
Stated accuracy	Margin 0.25 dB; on changes, a floor plus a fraction: power 0.25 dB and 15 per cent, OSNR 0.4 dB and 20 per cent, log bit error rate 0.3 decades and 20 per cent, DGD 1.0 ps and 20 per cent (`twin.yaml`)	Operations practice, consistent with published twin accuracies	A twin with structural error (arm A-full-cd-tm1)
Structural error	The world's GSNR about 0.55 dB below the twin's prediction (standard deviation 0.15 dB)	Stated assumption	Arm A-full-cd-tm1 on diagnosis cases 2000 to 2159, with and without the margin judged against the fleet

6.7 The ledger and scoring

Table 16. The ledger and scoring

Quantity	Value	Source	Test
Commitment threshold	0.80, with "none of these" below 0.20	Stated assumption, set on development cases	The warning test (`warnings`): raising it to 0.95 cuts wrong commitments from 49 to 17 of 1,686 but withholds 540 right or partly right
Equivalence	Two causes are one class when their signatures differ by at most two standard deviations	Stated assumption	Not tested
"Can this decide?"	Refuse a check of 0.5 hours or more with less than a 5 per cent chance of leading to a decision	Stated assumption, set on development cases	Not tested on the sealed cases
Pooled weights	General lines 1; specialised lines and return passes 0.25; a floor of 1 per cent spread over every named hypothesis	Stated assumption	Not tested

7. The sealed cases at a glance

Figure 2 draws the paper's Table III. It shows right diagnoses, hours on tests and travel, and site visits on the 320 sealed cases, for the three points of alarm-based troubleshooting and for the agent, each with status-quo instruments and with future instruments and every telemetry capability ("+ all"). Hours are field and remote hours together. The records are in `results/campaign/c4/` and the summary in `results/campaign/flagship_c4.json`.

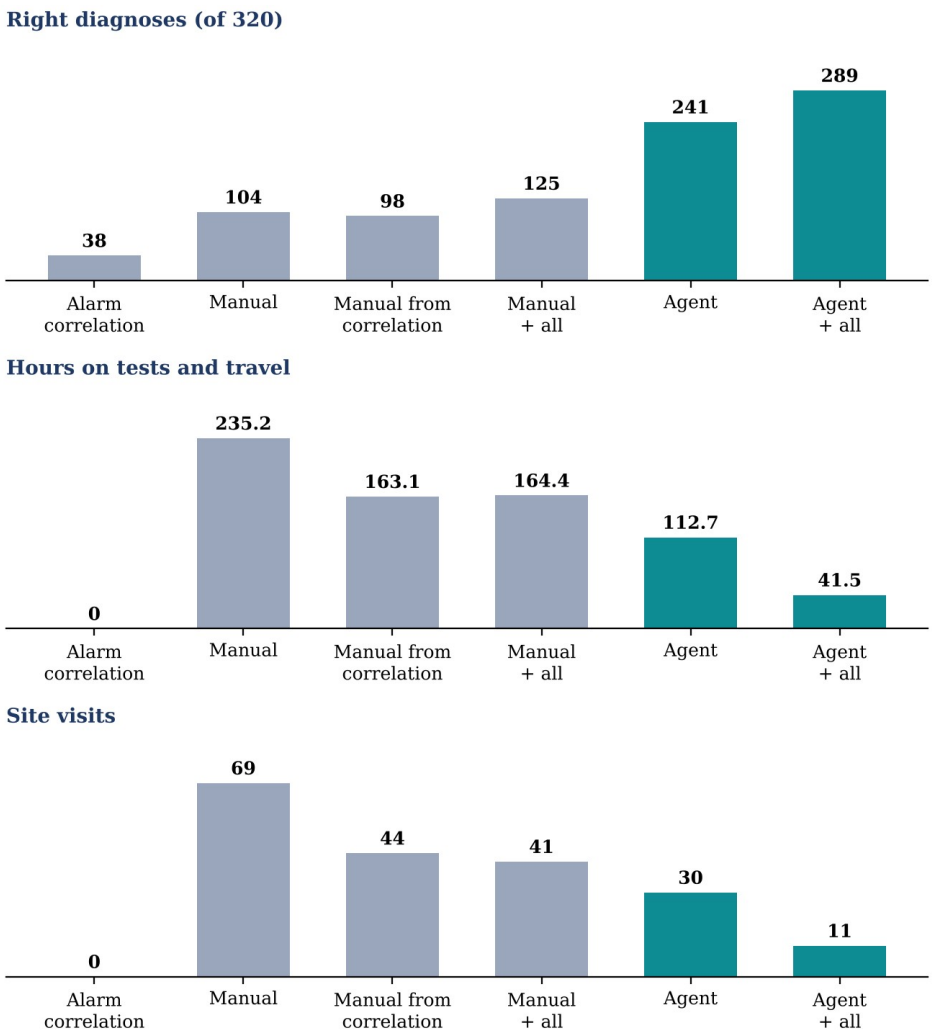


Figure 2. The sealed cases: right diagnoses, hours on tests and travel, and site visits. Grey: alarm correlation alone, manual troubleshooting, manual troubleshooting from correlation and manual troubleshooting with future instruments and every capability. Teal: the agent, as reported and with future instruments and every capability.

On the 102 cases that both the agent and manual troubleshooting get right, the agent spends 3.8 hours, all of them on remote tests, with no site visit. Manual troubleshooting spends 101.4 hours, 93.9 of them in the field, and makes 36 visits. The agent commits wrongly once in the 320 cases, and manual troubleshooting 19 times.

8. Reproduction

8.1 Installing and checking the code

The runs used Python 3.14 on a laptop with six worker processes; Python 3.11 or later suffices. From the root of the archive:

- `pip install -e ".[dev]"` installs the package and its dependencies: pydantic, numpy, scipy, networkx, pyyaml and the OpenAI client.

- `python scripts/exp2_freeze.py --check --manifest spec/CAMPAIGN_MANIFEST_C4V.json` confirms that every file of the agent, the simulator and the harness matches the manifest of the archived code. It prints “unchanged”.

The paper’s runs were made under three manifests. Manifest C4H covers every result except the test of a twin with wrong physics. Manifest C4T adds amendment 22, which carries the world’s structural mismatch to the network of each case’s day; the frozen agent’s runs of that test were made under it. Manifest C4V adds amendment 23, which lets a drift case be posed in a world that raises standing alarms (it changes no case reported in the paper), and amendment 24, the margin levels judged against the fleet, switched on by `ADAGENT_MARGIN_FLEET=1` and off by default. The archived code is C4V. It differs from C4H in four files (`src/adagent/world/history.py`, `src/adagent/harness/cases2.py`, `src/adagent/tools/margin.py` and `src/adagent/tools/loop.py`), and with the variable unset no change alters any case without the mismatch, which was confirmed case for case. The amendments are in `spec/memos/`.

- `python -m pytest -q tests` runs the tests. The full suite takes about a quarter of an hour.

The learned specialists ship as trained weights in `models/gnn/`, so nothing needs training. The fit cache that speeds up the twin is rebuilt on first use.

8.2 Running an arm

`scripts/exp2_run.py` runs one or more arms over a range of seeds and writes one record per case and arm, `s<seed>.json`, under the output folder. Its header lists the arms and options. For example:

- `python scripts/exp2_run.py --arms A-full-cd,B --seeds 2000-2019 --workers 4` reruns the agent as reported and manual troubleshooting on the first twenty diagnosis cases.

The arms of the paper are:

- **A-full-cd:** the agent as reported, with the “can this decide?” step.
- **A-full-cd-fut:** the same agent with future instruments.
- **A-Lk and A-Lall:** the same agent with one telemetry capability or with all of them.
- **B, B-fut, B-L1 and B-Lall:** manual troubleshooting under the same conditions.
- **B-ac and B-ac-fut:** manual troubleshooting started from correlation’s root alarm.
- **AC and AC-fut:** alarm correlation alone.
- **L-full-cd2:** the agent with the LLM orchestrator as its planner.
- **A-full-cd-lf:** the agent with the LLM framer.
- **LLM-1:** the LLM deciding alone.
- **A-full-cd-tm1 and B-tm1:** the agent and manual troubleshooting in a world with physics the twin does not model. With `ADAGENT_MARGIN_FLEET=1`, A-full-cd and A-full-cd-tm1 measure the margin levels judged against the fleet.

The sealed seeds are refused. `exp2_run.py` refuses seeds 7000 to 7319 unless the environment sets `ADAGENT_RECORD_RUN_C4=1`, as it was set once, for the record run of the frozen agent. Seeds 4000 to 4319, an earlier sealed set, are refused in the same way.

Running times. With six worker processes, an arm on 320 cases takes:

- a few minutes for alarm correlation alone;
- about half an hour for manual troubleshooting;
- about six hours for the agent;
- up to about ten hours for the agent with every telemetry capability.

Two-fault cases and duct cuts take longest, because the twin fits many candidates and their compounds.

8.3 What reproduces exactly

Every arm that calls no LLM reproduces exactly, case for case: the same seed gives the same network, fault, data, question, trajectory and score. One exception was found: in one case of the 160 checked (case 2133), a rerun chose a different check of the same kind on another transceiver and reached a different outcome. The likely cause is a tie between checks of equal value broken by the order of a set, which Python's string hashing varies between runs. Setting `PYTHONHASHSEED=0` fixes that order for reruns.

The LLM arms used gpt-5.6 through the OpenAI API under default sampling, with no temperature or seed set. A rerun needs an API key in the environment (`OPENAI_API_KEY`) and will differ in detail. The records of these arms hold each response of the LLM with each call's latency and its input, output and reasoning tokens. The prompts are not stored, because the code rebuilds each prompt from the case.

8.4 Where each result of the paper comes from

Table 17. The paper's results and their records

Result in the paper	Records
Table III, the sealed cases	`results/campaign/c4/`: A-full-cd, A-full-cd-fut, A-Lall, AC, AC-fut, B, B-fut, B-ac, B-ac-fut, B-Lall; summary in `results/campaign/flagship_c4.json`
Table IV, by kind of anomaly	The same records
Table V, the LLM deciding alone	`results/campaign/c4_dev/LLM-1`, against the agent (`results/campaign/c4r/A-full-cd`) and manual troubleshooting (`results/campaign/c4_dev/B`) on the diagnosis cases
Table VI, telemetry capabilities	`results/campaign/c4_dev/`: A-L1 to A-L6, A-Lall, B-L1, B-Lall, with A-full-cd-fut and B-fut as the baselines
Table VII, the LLM's roles	Planning: `results/campaign/c4/L-full-cd2` against A-full-cd. Proposing: `results/exp1_c4/`. Framing: `results/campaign/c4/A-full-cd-lf`. Reading the question: `results/campaign/qi19/`. Expressing the answer: `results/campaign/c4/ae17` and `results/c4h_cloud/ae_records`. Building the graph: `knowledge_pilot/out/`
A twin with wrong physics	The frozen agent and manual troubleshooting: `results/campaign/c4t/` (A-full-cd-tm1, B-tm1). The margin judged against the fleet: `results/campaign/c4v/` (A-full-cd, A-full-cd-tm1), each record holding the median offset removed (`margin_offset_decades`). The accurate-twin baselines are `results/campaign/c4r/A-full-cd` and `results/campaign/c4_dev/B`; `scripts/c4t_tm_summary.py` tabulates them
The degraded knowledge graph	`results/campaign/kgx_c4/`, with its control in `kgx_c4_control/`

Result in the paper	Records
Completing the documentation	First round: `results/campaign/a18/`. Second round, development and test: `results/campaign/a20/`, with the completions as `completion_*.json` and the trace of every changed case in `a20/knockouts_2026-11-04/`
The twin's measured error	`results/c4h_cloud/twin_fidelity.txt` and `twin_error_2026-11-04.txt`
Rules in the graph and in code	`results/c4h_cloud/rules_compare/`
Warning rules over 1,686 commitments	`results/campaign/warnings/`
The worked example, case 2021	`results/campaign/c4_trace/`
The mix of kinds reweighted	`results/campaign/mix_c4.json`

The run scripts that made these records are in `scripts/`:

- `run_c4_record.sh` and `run_c4*.sh`: the sealed cases and the campaign on the diagnosis cases;
- `run_a20_test.sh`: the test of documentation completion;
- `a19_qi_run.py`: question reading;
- `a17_ae_run.py`: answer expression;
- `run_c4t.sh`: the test of a twin with wrong physics (the frozen agent); the fleet-judged runs are `exp2_run.py` with `ADAGENT_MARGIN_FLEET=1`, as `spec/memos/AMENDMENT_24.md` records.

Each script states at its head what it runs, which manifest it checks and where its records go.